

SatQuery AI

AERIS — Agentic Earth Reasoning & Intelligence System

A Comprehensive Study & Technical Design Document

An Interactive Vision–Language Assistant for Multimodal Remote-Sensing
Image Analysis through Text Queries

Smart India Hackathon 2026 | Problem Statement **SIH26167** | ISRO

Document purpose. This document is simultaneously (i) a research survey of remote-sensing vision–language analysis, (ii) a technical system-design document, (iii) a feature specification, (iv) a team learning guide, (v) a development roadmap, and (vi) a risk-aware implementation blueprint for building SatQuery AI (working name of the intelligence layer: AERIS). It is written to be used, not merely read: every design claim is labelled as established fact, proposed architecture, assumption, experimental idea, or future work.

Abstract

SatQuery AI is an agentic Earth-observation intelligence platform that lets analysts interrogate satellite imagery in natural language. Instead of forcing users to choose sensors, bands, models, and GIS operations by hand, the system inspects the uploaded imagery, classifies the user’s intent, plans an execution graph, dispatches specialist remote-sensing models (detection, segmentation, spectral indices, change detection, optical–SAR fusion), grounds every claim in spatial evidence, and returns an auditable answer with confidence and a full execution trace. This document surveys the relevant literature (remote-sensing captioning, visual question answering, visual grounding, bi-temporal change detection, change description, optical–SAR fusion, and geospatial foundation models), catalogues the forty features proposed in the project’s concept notes, critically de-duplicates and prioritises them into MVP / SIH-demo / advanced / research tiers, specifies the end-to-end analysis pipeline and software architecture, and closes with an evaluation framework, a design-decision log, a professional risk register, a dependency-aware build order, and a team learning roadmap. The central engineering thesis is that *a vision-language model alone is not the product*: the defensible contribution is the orchestration layer that routes natural language to verified specialist computation and binds every answer to evidence.

How to read this document

Every non-trivial claim in this document carries one of five epistemic labels, following the critical-analysis requirement of the design brief:

- **[Fact]** — supported by peer-reviewed literature or official documentation (cited).
- **[Proposal]** — our engineering design for SatQuery AI; not yet validated.
- **[Assumption]** — plausible but must be validated during the build.
- **[Experimental]** — requires a prototype and measurement before adoption.
- **[Future]** — deliberately outside SIH scope.

Proposed features are never presented as existing capabilities.

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Part I

Foundations

1 Project Overview

1.1 What SatQuery AI is

SatQuery AI is an interactive vision–language assistant for multimodal remote-sensing image analysis through text queries. Its intelligence layer, AERIS (Agentic Earth Reasoning & Intelligence System), converts a natural-language request over one or more satellite images into a planned, executed, and audited geospatial analysis. The concept notes describe it as an *agentic Earth-observation intelligence platform* rather than a satellite chatbot: the differentiator is that answers are produced by verifiable specialist computation (spectral indices, segmentation, detection, change detection, optical–SAR fusion) and only *explained* by a language model — never invented by one.

Elevator pitch

You give SatQuery AI satellite imagery and ask questions in plain language — “*where did new buildings appear?*”, “*show me unhealthy vegetation*”, “*what changed between the optical and the SAR image?*” The system figures out what kind of data it is looking at, picks the right remote-sensing tools, runs them, highlights the exact regions that support its answer, and tells you how confident it is and how it got there. It turns satellite-image analysis from a specialist GIS workflow into a conversation — without hiding the evidence.

Technical definition [Proposal]. SatQuery AI is a deterministic-routing, tool-using multimodal analysis system: a query-understanding front end (classifier + planner) maps natural language and uploaded rasters to a typed execution graph over a registry of specialist models and geospatial operators; an evidence engine binds intermediate products (masks, boxes, index maps, change maps) to claims; and a vision–language model (VLM) performs perception-level tasks (VQA, captioning, grounding) and synthesises explanations strictly from structured, validated results.

1.2 The SIH problem statement

The project targets Smart India Hackathon 2026 problem statement **SIH26167** (ISRO): an interactive vision–language assistant for multimodal remote-sensing image analysis through text queries. The statement’s core demands map onto four system obligations: (i) accept *multimodal* Earth-observation input (at minimum optical and SAR); (ii) accept *text queries* as the primary interaction mode; (iii) perform *analysis*, not just description — detection, localisation, comparison, quantification; and (iv) remain *interactive* — follow-up questions, region-scoped queries, and iterative investigation.

1.3 The core problem being solved

Operationally, remote-sensing analysis today is a toolchain problem. Answering even a simple question such as “*did built-up area increase here?*” requires an analyst to know which sensor is appropriate, acquire and preprocess the data (atmospheric correction, cloud masking, co-registration), choose and run a change-detection method, post-process the result, quantify area, and interpret it. Each step has its own software, parameters, and failure modes. The bottleneck is not the availability of algorithms — it is the *selection, sequencing, and interpretation* of algorithms. SatQuery AI attacks exactly that bottleneck.

1.4 Why conventional remote-sensing analysis is difficult

- **Data heterogeneity [Fact].** Optical multispectral, SAR, thermal, and hyperspectral products differ in geometry, radiometry, noise model, and semantics; a SAR backscatter image cannot be treated as an RGB picture (Section 3).
- **Preprocessing burden [Fact].** Georeferencing, CRS handling, resampling, co-registration, cloud masking, and atmospheric correction are prerequisites for almost any analysis and are a common source of silent error.
- **Scale [Fact].** A single Sentinel-2 tile is roughly $10,980 \times 10,980$ pixels per band at 10 m; naive full-scene inference is infeasible, so tiling and spatial reconstruction are mandatory.
- **Expertise gap.** The people with the questions (disaster responders, planners, agricultural officers) are usually not the people who can operate the toolchain.

1.5 Why multimodal satellite imagery is difficult to analyse

Optical and SAR are *complementary but non-aligned* observations of the same ground truth. Optical sensors measure reflected solar radiation in spectral bands; SAR measures coherent microwave backscatter sensitive to structure, moisture, and geometry, works at night and through clouds, but suffers speckle, layover, foreshortening, and shadow [Fact; Section 3]. Fusing them requires radiometric and geometric normalisation, co-registration to sub-pixel accuracy, and models that respect that optical $\neq$ SAR at the pixel-interpretation level.

1.6 Why natural-language interaction is useful

Natural language is the shortest path between an analyst’s question and the system’s capabilities. It removes the requirement to know *which* tool implements *which* analysis, allows iterative refinement (“now only the north-east part”), and produces a self-documenting session: the query log *is* the analysis narrative. It does not remove the need for rigour — which is why the language layer must sit *on top of* verifiable computation, not replace it.

1.7 Why a VLM alone is insufficient

Critical analysis — the core argument

A generic or even remote-sensing-adapted VLM cannot be the whole system, for five reasons:

1. **Hallucination without evidence [Fact].** Generative models produce fluent answers unsupported by the pixels; without grounding, there is no way to audit a claim.
2. **No quantitative rigour [Fact].** “How many hectares changed?” requires georeferenced pixel counting, not token prediction. VLMs are poor at precise counting, measurement, and spatial arithmetic.
3. **Band blindness [Fact].** Standard VLMs consume 3-channel RGB. NDVI, NDWI, SAR polarimetry, and 13-band Sentinel-2 products are invisible to them unless specialist preprocessing renders or computes them.
4. **Weak temporal/cross-modal reasoning [Fact].** Bi-temporal change reasoning and optical–SAR comparison are active research problems (Sections 7, 8, 9); off-the-shelf VLMs do not solve them reliably.
5. **No provenance [Fact].** A chat answer carries no trace of inputs, parameters, model versions, or intermediate products — unacceptable for scientific or operational use.

Therefore: **the VLM is a component, not the architecture.** Specialist models produce structured, checkable results; the VLM perceives and explains.

1.8 Why specialist remote-sensing tools/models are required

Indices (NDVI, NDWI, NDBI, NBR) are closed-form band arithmetic with decades of validation; segmentation and detection models (U-Net, SegFormer, task-specific detectors) produce pixel-accurate, measurable outputs; change-detection networks (Siamese/transformer/Mamba families) are purpose-built for bi-temporal comparison. These are deterministic or statistically evaluated tools with known error characteristics — exactly what an evidence-first system needs as its ground truth layer.

1.9 What makes this different from a satellite-image chatbot

- **Evidence-first answers:** every claim links to a region, mask, model, and execution trace (Section 20).
- **Tool routing, not monolithic generation:** a planner dispatches specialist models (Section 12).
- **Multimodal and temporal by design:** optical, SAR, and bi-temporal inputs are first-class, with compatibility validation.
- **Quantified output:** areas, percentages, counts — computed geospatially.
- **Auditable:** provenance, confidence, and reproducibility are system features, not afterthoughts (Section 21).

1.10 Intended end-to-end user experience

Upload imagery → ask a question → watch the system state its plan → receive an answer with highlighted evidence, confidence, and a trace → drill in (“investigate”, “ask this region”, “show me”) → export a report. The concept notes’ scripted dialogue (“*built-up area increased by approximately 18.4%, 14.2 hectares, primarily north-east, confidence 91%*”) is the target interaction quality.

1.11 Novelty and technical contribution

Novelty is claimed *only* at the integration level [Proposal]: combining natural-language routing + specialist remote-sensing tools + evidence grounding + temporal and cross-modal reasoning + auditable traces in one SIH-feasible system. Each individual ingredient exists in the literature (Section 28); the combination, and the engineering discipline around evidence, is the contribution. We explicitly do *not* claim a new foundation model or a new algorithm.

1.12 What should realistically be demonstrated in an SIH prototype

SIH demo target (reasoned in Section 23)

A judge should be able to: upload an optical GeoTIFF and a SAR image, ask single-image questions (VQA, captioning, grounding, indices), run a bi-temporal change query with a quantified, localised answer, see cross-modal evidence, inspect the execution trace, and export a report — live, in minutes, on a single GPU. Continuous monitoring, voice control, 3D Earth, and autonomous multi-step investigation are explicitly *not* required for a credible demo.

2 Who the User Is and What They Need

2.1 Target users and workflows

Table 1 characterises the intended users. The common pattern: users have *questions*, not *pipelines*; they need localised, quantified, trustworthy answers faster than the manual toolchain delivers them.

Table 1: Target users, their data, questions, pain points, and required outputs.

User	Data	Typical questions	Current difficulty	Needed output / expertise
Remote-sensing analyst	S-1/S-2, Landsat, VHR	Which model for this task? Did preprocessing succeed?	Toolchain fragmentation; repetitive scripting	Scripted-quality results without scripting; high expertise
ISRO / scientific researcher	Cartosat, RISAT, EOS series	Does this method generalise? What is the error?	Reproducibility, benchmarking overhead	Auditable traces, metrics; very high expertise
Disaster-management analyst	SAR + optical, rapid tasking	What is damaged? Where is flooding?	Cloud cover blocks optical; time pressure	Fast change maps, confidence; medium expertise
Agriculture analyst	S-2, Landsat time series	Which fields are stressed? When?	Index computation + zonal stats by hand	NDVI/health maps, alerts; medium expertise
Urban planner	VHR optical	Where is new construction? Is it legal?	Manual comparison of dated imagery	Change masks, area stats; low-medium expertise
Environmental researcher	Multispectral time series	Water spread? Vegetation loss?	Long time-series handling	Temporal summaries, trends; medium expertise
Defence / intelligence analyst	VHR optical + SAR	What changed at this site? Is this new?	Cross-sensor correlation; audit requirements	Evidence-linked findings; high expertise
GIS professional	Mixed raster/vector	Integrate EO results into GIS	Format/CRS friction	GeoJSON/GeoTIFF exports; high expertise
Student / researcher	Public datasets	How do I reproduce this?	Steep learning curve	Interactive exploration; low expertise

2.2 User → Need → Query → Analysis → Output

Table 2 is the canonical mapping that drives feature design. Each row is an end-to-end contract the system must satisfy.

Table 2: User → Need → Query → Analysis → Output mapping.

User	Need	Example query	Analysis path	Output
Planner	Detect new construction	“Where did new buildings appear?”	Registration → change detection → building segmentation → intersection	Change mask + area + evidence
Disaster analyst	Rapid damage view	“Compare these two images and tell me what changed.”	Bi-temporal change detection + change classification	Change map + summary + confidence
Agriculture analyst	Crop stress	“Show me areas with unhealthy vegetation.”	Band check → NDVI → threshold/zones	NDVI map + stressed-region mask
Environmental researcher	Water monitoring	“Where are the water bodies?”	NDWI/MNDWI + water segmentation	Water mask + area
Analyst	Scene understanding	“What land-cover types are visible here?”	Scene classification / land-cover segmentation	Class map + proportions
Analyst	Cross-sensor check	“What changed between the optical and SAR images?”	Modality validation → per-sensor analysis → joint interpretation	Comparative explanation + dual evidence
Researcher	Spectral query	“Which regions have high NDVI?”	Index engine → threshold → zonal stats	Thresholded map + stats
Analyst	Infrastructure	“Find all large roads in this region.”	Detection/segmentation → vectorisation → size filter	Boxes/masks + list
Any user	Trust	“Show the region responsible for your answer.”	Evidence engine → grounding link	Highlighted region + trace
Any user	Interpretation	“Explain why this area appears different.”	Evidence aggregation → VLM explanation	Grounded explanation

2.3 Representative natural-language queries

The system must handle at least the following query families (used later as routing test cases in Section 19): land-cover identification; water-body localisation; vegetation-health mapping; bi-temporal comparison; construction detection; optical–SAR comparison; linear-infrastructure extraction; index thresholding; causal explanation; and evidence recall (“show the region responsible for your answer”). Each family exercises a different routing path, which is why the query set doubles as an integration test suite [Proposal].

3 Remote Sensing Fundamentals Required for the Project

This section is the study baseline: every team member must be fluent in Section 3.1; specialists go deeper per the learning matrix (Section 17).

3.1 Remote-sensing fundamentals

3.1.1 Electromagnetic spectrum and sensing modes

Remote sensing measures electromagnetic (EM) radiation interacting with the Earth’s surface. *Passive* sensors record reflected sunlight or emitted thermal radiation (optical, multispectral, hyperspectral, thermal); *active* sensors emit their own signal and record the return (SAR, LiDAR)

[Fact]. Key spectral regions for this project: visible ($\sim 0.4\text{--}0.7\ \mu\text{m}$), near-infrared (NIR, $\sim 0.7\text{--}1.3\ \mu\text{m}$), short-wave infrared (SWIR, $\sim 1.3\text{--}3\ \mu\text{m}$), thermal infrared ($\sim 8\text{--}14\ \mu\text{m}$), and microwave (e.g. C-band, $\sim 5.6\ \text{cm}$ for Sentinel-1).

3.1.2 The four resolutions

- **Spatial resolution** — ground sampling distance (GSD): the ground size of one pixel (Sentinel-2: 10/20/60 m; Landsat 8/9: 30 m multispectral; MODIS: 250–1000 m). Determines the smallest detectable object.
- **Spectral resolution** — number and width of bands. Multispectral: few broad bands (Sentinel-2: 13). Hyperspectral: hundreds of narrow contiguous bands.
- **Temporal resolution** — revisit time (Sentinel-2: ~ 5 days with two satellites; Landsat: 16 days; MODIS: daily). Governs change-analysis cadence.
- **Radiometric resolution** — quantisation depth / sensitivity to intensity differences (Sentinel-2: 12-bit). Affects index precision and subtle-change detection.

3.1.3 Multispectral, hyperspectral, optical, thermal, SAR

Optical/multispectral imagery records reflected solar radiation in discrete bands — intuitive, rich in spectral semantics (vegetation, water, soil), but blocked by clouds and darkness. *Hyperspectral* adds fine spectral discrimination (minerals, species, materials) at high data and processing cost. *Thermal* records emitted radiation (surface temperature, urban heat, fires). *SAR* (synthetic aperture radar) actively illuminates the surface with microwaves and records backscatter amplitude and phase; Sentinel-1 provides C-band VV/VH polarisations. SAR penetrates clouds and works at night and is sensitive to structure, roughness, and dielectric properties (moisture), but exhibits speckle (multiplicative granular noise), and geometric distortions: *layover* (tops of structures mapped toward the sensor), *foreshortening* (slopes compressed), and *radar shadow* (no-return zones) [Fact]. Consequence for SatQuery AI: SAR outputs must never be interpreted with optical intuition; the explanation layer needs modality-aware phrasing [Proposal].

3.2 Important satellite data concepts

3.2.1 Missions relevant to SatQuery AI

- **Sentinel-1 (ESA/Copernicus)** — C-band SAR, VV/VH (IW mode), $\sim 10\ \text{m}$ pixel spacing, 6–12 day revisit; the project’s primary SAR source.
- **Sentinel-2 (ESA/Copernicus)** — 13-band multispectral, 10–60 m, ~ 5 -day revisit; primary optical source; L2A products are atmospherically corrected (Sen2Cor) and include a scene classification (cloud) layer.
- **Landsat 8/9 (USGS/NASA)** — 30 m multispectral + 15 m panchromatic + thermal; 16-day revisit; 50+ year archive — the reference for long-term change.
- **MODIS (NASA)** — 250–1000 m, daily; coarse but temporally dense.
- **VIIRS (NOAA/NASA)** — 375–750 m, daily; night-lights band relevant to urbanisation.
- **ISRO datasets** — Cartosat series (high-resolution panchromatic/multispectral, e.g. Cartosat-3 $\sim 0.25\ \text{m}$ pan), RISAT series (C-band SAR), Resourcesat (LISS-III/IV), EOS-04 (RISAT-1A). Accessible via the Bhuvan geoportal and MOSDAC; licensing and bulk-access terms must be verified before training use [Assumption].

3.2.2 Data artefacts the system must understand

- **Bands and metadata:** band order/wavelengths, acquisition time, sensor mode, processing level, nodata value, scale/offset.

- **Georeferencing and CRS:** the affine geotransform mapping pixel (*col, row*) to projected coordinates; CRS identified by EPSG code (e.g. WGS84 = EPSG:4326; UTM zones = EPSG:326xx/327xx). Mixing CRSs without reprojection is a silent-killer bug class (Section 24).
- **Formats:** GeoTIFF (georeferenced TIFF, the workhorse), JP2 (JPEG-2000; Sentinel-2 distribution format), plus PNG/JPEG for benchmark datasets (no georeferencing).
- **Raster tiles:** large scenes are processed in windows (e.g. 512×512 with overlap) and spatially reconstructed.
- **Cloud masking:** Sentinel-2 L2A SCL layer or s2cloudless; clouds/shadows must be masked or flagged before indices and change detection.
- **Atmospheric correction:** converting top-of-atmosphere reflectance to bottom-of-atmosphere (surface) reflectance (L1C $\rightarrow$ L2A); indices computed on L1C data are biased [Fact].
- **Resampling:** bringing bands/frames to a common grid (nearest for labels, bilinear/cubic for continuous data).
- **Co-registration:** aligning two scenes to sub-pixel agreement; mandatory before any bi-temporal or cross-modal comparison (Section 7).

3.3 Remote-sensing indices

Indices are the cheapest, most defensible “AI” in the system: closed-form band arithmetic with physical meaning. For each index we give **Formula** $\rightarrow$ **Bands** $\rightarrow$ **Meaning** $\rightarrow$ **Typical query** $\rightarrow$ **Output** $\rightarrow$ **Limitations**. All formulas assume surface reflectance ρ in the named bands.

3.3.1 NDVI — Normalized Difference Vegetation Index [1]

$$\text{NDVI} = \frac{\rho_{\text{NIR}} - \rho_{\text{Red}}}{\rho_{\text{NIR}} + \rho_{\text{Red}}} \quad \text{Sentinel-2: } \frac{B08 - B04}{B08 + B04} \quad (1)$$

Bands: NIR, Red. **Meaning:** healthy vegetation absorbs red and reflects NIR strongly; NDVI $\in [-1, 1]$, dense healthy vegetation typically > 0.4 , water < 0 . **Query:** “Show me areas with unhealthy vegetation.” **Output:** index map + thresholded stress mask. **Limitations:** saturates over dense canopy; sensitive to soil background and atmosphere; not directly comparable across sensors without calibration.

3.3.2 EVI — Enhanced Vegetation Index [3]

$$\text{EVI} = G \frac{\rho_{\text{NIR}} - \rho_{\text{Red}}}{\rho_{\text{NIR}} + C_1 \rho_{\text{Red}} - C_2 \rho_{\text{Blue}} + L}, \quad G=2.5, C_1=6, C_2=7.5, L=1 \quad (2)$$

Bands: NIR, Red, Blue. **Meaning:** like NDVI but corrects aerosol and soil effects and resists saturation in high-biomass areas. **Query:** “Compare vegetation vigour between these dates.” **Output:** EVI map/difference. **Limitations:** needs blue band; coefficients are empirical (MODIS-derived); can be noisy in arid regions.

3.3.3 SAVI — Soil-Adjusted Vegetation Index [2]

$$\text{SAVI} = (1 + L) \frac{\rho_{\text{NIR}} - \rho_{\text{Red}}}{\rho_{\text{NIR}} + \rho_{\text{Red}} + L}, \quad L = 0.5 \text{ (typical)} \quad (3)$$

Bands: NIR, Red. **Meaning:** NDVI with a soil-brightness correction factor L ; better in sparse vegetation / exposed soil. **Query:** “Vegetation cover in this semi-arid region?” **Output:** SAVI map. **Limitations:** L is scene-dependent; still saturates at high biomass.

3.3.4 NDWI — Normalized Difference Water Index [4]

$$\text{NDWI} = \frac{\rho_{\text{Green}} - \rho_{\text{NIR}}}{\rho_{\text{Green}} + \rho_{\text{NIR}}} \quad \text{Sentinel-2: } \frac{B03 - B08}{B03 + B08} \quad (4)$$

Bands: Green, NIR. **Meaning:** open water reflects green and absorbs NIR; positive values indicate water. **Query:** “Where are the water bodies?” **Output:** water mask + area. **Limitations:** confounded by built-up land and wet soil; McFeeters NDWI is not ideal over mixed urban/water scenes.

3.3.5 MNDWI — Modified NDWI [5]

$$\text{MNDWI} = \frac{\rho_{\text{Green}} - \rho_{\text{SWIR}}}{\rho_{\text{Green}} + \rho_{\text{SWIR}}} \quad \text{Sentinel-2: } \frac{B03 - B11}{B03 + B11} \quad (5)$$

Bands: Green, SWIR. **Meaning:** suppresses built-up noise; the standard choice for water extraction in urbanised scenes. **Query:** “Map flood extent.” **Output:** robust water mask. **Limitations:** needs SWIR (20 m on Sentinel-2 → resampling); turbid/shallow water can be ambiguous.

3.3.6 NDBI — Normalized Difference Built-up Index [6]

$$\text{NDBI} = \frac{\rho_{\text{SWIR}} - \rho_{\text{NIR}}}{\rho_{\text{SWIR}} + \rho_{\text{NIR}}} \quad \text{Sentinel-2: } \frac{B11 - B08}{B11 + B08} \quad (6)$$

Bands: SWIR, NIR. **Meaning:** built-up areas reflect SWIR more than NIR — the spectral mirror of NDVI. **Query:** “Show built-up regions.” **Output:** built-up likelihood map. **Limitations:** confuses bare soil with buildings; use as a feature/prior, not a sole detector [Proposal].

3.3.7 NBR — Normalized Burn Ratio [7]

$$\text{NBR} = \frac{\rho_{\text{NIR}} - \rho_{\text{SWIR2}}}{\rho_{\text{NIR}} + \rho_{\text{SWIR2}}} \quad \text{Sentinel-2: } \frac{B08 - B12}{B08 + B12}, \quad \Delta\text{NBR} = \text{NBR}_{\text{pre}} - \text{NBR}_{\text{post}} \quad (7)$$

Bands: NIR, SWIR2 (B12). **Meaning:** burned areas show strong SWIR2 response; ΔNBR quantifies burn severity. **Query:** “Assess fire damage in this region.” **Output:** burn-severity map. **Limitations:** seasonal/illumination differences mimic severity; needs co-registered pre/post pair.

Design consequence — the spectral-index engine

Indices require *band identification* (from metadata or user hint), *surface reflectance* (warn if L1C), *cloud masking*, and *resolution harmonisation* (SWIR at 20 m must be resampled to the analysis grid). The index engine is therefore a module with validation, not a set of formulas (Section 16). Notably, the concept notes never mention an index engine even though user queries (“high NDVI”, “unhealthy vegetation”) demand one — this is a genuine gap the architecture fills [Proposal; see Section 13.4].

Part II

Research Survey

4 Previous Research: Remote-Sensing Image Captioning

4.1 Evolution of the task

Remote-sensing image captioning (RSIC) has passed through five technical generations [Fact]:

1. **CNN–RNN encoder–decoder** (mid-2010s): a CNN encodes the image; an LSTM decodes a sentence, following the “show and tell” paradigm. Early RS adaptations produced short, generic descriptions (“many buildings are near a road”).
2. **Attention-based captioning**: soft attention lets the decoder focus on image regions while generating words, improving object mention and reducing generic output.
3. **Transformer-based systems**: self-attention encoders/decoders [8] replaced recurrence, enabling longer-range context and better fluent, structured descriptions.
4. **Multimodal pretraining**: CLIP-style contrastive image–text alignment [11] provided transferable features; captioning became fine-tuning on top of aligned encoders.
5. **Large vision–language models**: instruction-tuned LVLMs (LLaVA-style [13]) adapted to RS — RSGPT [24], RS-LLaVA [26], GeoChat [25], SkyEyeGPT [27], LHRs-Bot [28], EarthGPT [29] — unify captioning with VQA and grounding in one model.

4.2 From generic description to geospatial semantics

Progress has moved descriptions along five axes: (i) *land-cover semantics* (correct class vocabulary); (ii) *geographic semantics* (place-aware phrasing); (iii) *object relationships* (“harbour with docked ships”); (iv) *spatial descriptions* (“in the north-east corner”); and (v) *domain terminology* (“built-up”, “cropland”, “sediment plume”). Benchmarks such as UCM-Captions, Sydney-Captions, RSICD, NWPU-Captions, and the human-annotated RSICap [24] drove this progression; VRSBench [23] now evaluates captioning jointly with VQA and grounding.

4.3 Limitations of generic VLMs on satellite imagery

Why not just use GPT-4V-class or LLaVA-class models directly?

- **Top-down domain shift [Fact]**: nadir views, tiny objects, unusual textures, and absence of canonical object scales degrade encoders trained on ground-level photos.
- **Resolution mismatch [Fact]**: RS scenes are megapixels; VLM vision encoders expect ~224–448 px inputs — naive downsampling destroys exactly the detail the question targets.
- **Weak spatial reasoning [Fact]**: VLMs are unreliable at counting, relative position, and area estimation in overhead imagery; benchmarks repeatedly show large gaps versus specialised models.
- **No spectral access [Fact]**: RGB-only input discards NIR/SWIR/SAR information that defines the semantics of RS scenes.

Domain-adapted RS-VLMs close part of this gap; SatQuery’s routing + specialist tools close the rest [Proposal].

5 Remote-Sensing Visual Question Answering (RSVQA)

5.1 Task formulation

RSVQA formalises: *input image + natural-language question* → *model reasoning* → *answer*. The seminal RSVQA dataset and baselines [21] introduced low- and high-resolution splits with questions about presence, count, comparison, and area; HRVQA [22] extended this to very high-resolution aerial imagery. VRSBench [23] broadened evaluation to captioning, grounding, and VQA with human-verified annotations.

5.2 Methods

Three families dominate [Fact]: (i) *fusion baselines* — CNN/ViT image features fused with question embeddings (the original RSVQA baselines); (ii) *domain-adapted VLMs* — instruction-tuned

LVLMS (RSGPT [24], GeoChat [25], RS-LLaVA [26], LHRS-Bot [28], SkyEyeGPT [27], EarthGPT [29]); and (iii) *contrastively-aligned encoders* (RemoteCLIP [30], GeoRSCLIP/RS5M [31]) used with answer heads or retrieval. Instruction tuning on RS image–text pairs is the common enabler; vision–language alignment quality largely determines VQA accuracy.

5.3 Open difficulties

Spatial reasoning (relative position, orientation, distance), *object counting* (especially small dense objects), and *scene understanding* at multiple scales remain weak points even for RS-adapted VLMs [Fact]. Quantitative questions (“how many hectares”) are fundamentally unsuited to token generation.

5.4 What existing systems solve — and what they do not

Table 3: RSVQA/RS-VLM capability boundary relevant to SatQuery AI.

Solved reasonably well	Not solved / unreliable
Presence/absence questions; scene-level description; coarse land-cover vocabulary; single-image VQA on benchmark-like data	Precise counting and measurement; reliable localisation of the answer’s evidence; bi-temporal and cross-modal (optical–SAR) reasoning; quantitative area/percentage answers; calibrated confidence; provenance

SatQuery AI accepts VLM-class answers for perception-level questions but routes quantitative, temporal, and cross-modal questions to specialist tools whose outputs are measurable [Proposal]. This division of labour is the system’s central design decision (Section 23).

6 Text-Guided Region Grounding in Remote Sensing

6.1 Task family

Referring expressions describe a target (“the large water body in the north”); *referring-expression comprehension (REC)* returns a bounding box; *referring-expression segmentation (RES)* returns a pixel mask; *phrase grounding* localises all mentioned entities; *point grounding* returns a representative point. Outputs range over region proposals, boxes, masks, and points. General-domain enablers include Grounding DINO [15] (open-vocabulary box grounding) and SAM [14] (promptable masks).

6.2 Remote-sensing grounding resources

RS-specific grounding datasets include RSVG [32], DIOR-RSVG [33], RRSIS-D [34] (mask-level), and the grounding splits of GeoChat [25] and VRSBench [23]. RS-VLMs with grounding heads (GeoChat, LHRS-Bot variants, SkyEyeGPT) output boxes or points inline with conversation.

6.3 Answer generation vs. evidence localisation

The distinction that defines SatQuery AI

Saying “*there is a water body*” is **answer generation**. Answering “*where is the large water body?*” with a highlighted polygon is **evidence localisation**. Grounding converts the VLM from an oracle into an instrument: every claim can be checked against pixels. In SatQuery AI, grounding is not only a user-facing feature — it is the mechanism that binds specialist-model outputs (masks, boxes, change regions) to language in the evidence layer (Section 20) [Proposal].

For the SIH build, box-level grounding via an RS-VLM (GeoChat-class) plus mask-level grounding from specialist segmentation models is the pragmatic combination; end-to-end text-to-mask RES

on arbitrary RS scenes is research-grade and deferred (Section 25).

7 Bi-Temporal Change Detection

7.1 Task and prerequisites

Bi-temporal change detection takes co-registered images I_{T0}, I_{T1} and produces a change map — binary (changed/unchanged) or semantic (from-class $\rightarrow$ to-class). Two prerequisites dominate all downstream quality [Fact]:

- **Image registration:** geometric alignment via georeferencing + (if needed) feature-based refinement (affine/homography); misregistration of even 1–2 pixels creates systematic false change along edges.
- **Radiometric normalisation:** season, illumination, and atmosphere differences mimic change; use same-season pairs, surface reflectance, and cloud masking.

7.2 Optical–optical change detection

Canonical use cases: Sentinel-2 $\rightarrow$ Sentinel-2 seasonal comparison, construction monitoring, vegetation loss, water expansion, disaster damage. The architectural evolution [Fact]:

1. **CNN differencing / early fusion:** concatenate or difference bi-temporal features; CVA-style baselines.
2. **Siamese CNNs:** weight-shared encoders with late fusion — FC-EF, FC-Siam-conc, FC-Siam-diff [35]; SNUNet adds dense skip connections [40]; DSIFN adds deep supervision [41].
3. **U-Net decoders:** Siamese encoders + U-Net-style decoders became the default dense-prediction recipe [19].
4. **Transformers:** BIT tokenises bi-temporal features and models context with a transformer [38]; ChangeFormer hierarchical transformer backbones [39]; STANet introduced spatial-temporal attention with the LEVIR-CD dataset [37].
5. **Hybrid and state-space models:** CNN–transformer hybrids; Mamba/state-space approaches (ChangeMamba’s MambaBCD/SCD/BDA [42], building on Mamba [18]) target global context at linear cost.

Benchmarks: LEVIR-CD [37], WHU-CD [43], OSCD [36], S2Looking [44], SYSU-CD [45], and the semantic-change dataset SECOND [46].

7.3 Optical–SAR change detection

Optical–SAR pairs (Sentinel-1 + Sentinel-2) combine *spectral* with *structural* information and enable cloud-resistant monitoring. Difficulties [Fact]: (i) heterogeneous appearance — the same object looks unrelated across modalities, so pixel-level differencing is meaningless; comparison must happen in feature/semantic space; (ii) cross-modal alignment — radiometric and geometric normalisation plus co-registration; (iii) fusion strategy — per-modality feature extraction followed by feature-level or decision-level change reasoning. Multimodal change datasets remain scarce relative to optical–optical, which is why SatQuery AI treats optical–SAR change as *comparative evidence* (per-sensor analysis + joint interpretation) rather than a single learned cross-modal change network for the SIH build [Proposal; see Section 9].

7.4 Pipeline anatomy and failure modes

registration $\rightarrow$ feature extraction $\rightarrow$ difference representation $\rightarrow$ change map $\rightarrow$ post-processing

Post-processing: morphological cleanup, minimum-area filtering, vectorisation. Dominant false-positive sources: misregistration edges, seasonal vegetation cycles, cloud/shadow residuals, and

sensor/angle differences. SatQuery AI mitigations: registration-quality check in the validation stage, cloud masking, seasonal-pair warnings, and change-confidence maps [Proposal].

8 Change Description and Change-Based VQA

8.1 Beyond binary masks

A change mask answers *where*; users ask *what, when, how much, and why*. The research ladder [Fact]:

Change Detection → Change Localization → Change Classification → Natural-Language Change I

Change captioning generates sentences describing the difference (LEVIR-CC and dual-branch transformer captioners [47]); *temporal/change VQA* answers questions over image pairs — CDVQA, the first benchmark for the task, provides 2,968 SECOND-based image pairs and >122k question-answer pairs with a Siamese baseline [48]; *change QA + grounding* (CDQAG task, QAG-360K dataset, VisTA baseline) couples textual answers with change masks [49]; instruction-tuned temporal VLMs (ChangeChat-87k [50]; TEOChat for temporal EO sequences [51]) extend this to multi-turn dialogue.

8.2 Question types the system must support

“What changed?” (classification), “where did it change?” (localisation + grounding), “when did it change?” (requires >2 observations or external dating [Future]), “how significant is the change?” (quantification: area, percentage — computed geospatially, not generated), and comparative/maximal queries (“largest change”).

Critical analysis — SatQuery AI’s change stack

The concept notes list change detection, segmentation, classification, description, and VQA as parallel features. In reality they form a dependency chain: description and VQA are only trustworthy when they verbalise *validated* masks and statistics. SatQuery AI therefore implements change description/VQA as *structured-results-to-language* rendering over the change pipeline’s outputs, optionally cross-checked by a change-VQA model — not as free-form VLM generation [Proposal]. This directly implements the evidence-first principle and matches the direction of CDQAG-style research [49].

9 Optical–SAR Multimodal Fusion

9.1 Fusion strategies

- **Early (data-level) fusion:** stack co-registered channels (e.g. S-2 bands + VV/VH) and train one encoder. Simple; sensitive to misregistration and radiometric mismatch; supported by paired datasets such as SEN12MS [52] and BigEarthNet-MM [53].
- **Feature-level (intermediate) fusion:** modality-specific encoders; merge features via concatenation, addition, or gating. The default for heterogeneous sensors.
- **Late (decision-level) fusion:** independent per-modality predictions combined by rules or a meta-model. Most robust to missing/degraded modalities — and the most auditable, since each modality’s evidence stays separable.
- **Cross-attention fusion:** queries from one modality attend to the other’s features; used in recent multimodal transformers and fusion-oriented foundation models (CROMA [59], DOFA [60], SkySense [61]).
- **Semantic-prior-guided / frequency-aware fusion / mixture-of-experts:** research directions that route or weight modality contributions by learned priors, frequency content,

or expert gating [Experimental for SatQuery AI].

9.2 When to fuse and when not to

Fusion policy for SatQuery AI (Proposal)

Fuse at **feature or decision level**, never raw-pixel level. Use fusion when tasks benefit from complementarity: built-up/water mapping under clouds, flood mapping (SAR water + optical context), structural confirmation of optical change. Keep modalities *separate* when: co-registration quality is poor; the question is purely spectral (NDVI) or purely structural (SAR texture); or auditability demands per-sensor evidence. SatQuery AI's default is *late fusion with joint explanation*: optical evidence + SAR evidence → fused conclusion, with each side inspectable — mirroring the concept notes' Cross-Modal Analysis Lab.

10 Multimodal Geospatial Foundation Models

10.1 Pretraining paradigms

Three paradigms underpin RS foundation models [Fact]: *contrastive pretraining* (CLIP-style image–text alignment: RemoteCLIP [30], GeoRSCLIP/RS5M [31]); *masked image modelling* (MAE-style [10] adapted to RS: SatMAE [55], Scale-MAE [56], SpectralGPT [57], Prithvi [58]); and *instruction tuning* of LVLMs on RS data (Section 5). Adaptation to downstream tasks uses parameter-efficient fine-tuning — LoRA [16], QLoRA [17], adapters — rather than full fine-tuning, which is compute-prohibitive for an SIH team.

10.2 Model landscape and SatQuery AI roles

Table 4 evaluates candidates on RS compatibility, multimodality, resolution handling, inference cost, openness, fine-tunability, spatial reasoning, and deployment feasibility. No model is recommended on popularity; each is assigned a concrete system role or rejected.

Table 4: Foundation-model / VLM comparison for SatQuery AI (assessment: [Proposal] unless cited).

Model / family	Input	Output	RS-specific?	Strength	Limitation	Possible Sat-Query AI role
CLIP [11]	RGB image + text	embeddings	No	transferable features	featur ground-level domain shift	baseline only
RemoteCLIP [30]	RGB RS + text	embeddings	Yes	RS-aligned retrieval/zero-shot	RGB-only; no generation	embeddings for retrieval/classification
GeoRSCLIP (RS5M) [31]	RGB RS + text	embeddings	Yes	large-scale alignment	RS RGB-only	embedding backbone candidate
LLaVA-1.5 [13]	RGB + text	text	No	strong VQA	general poor RS semantics	generic fallback / comparison baseline
GeoChat [25]	RGB RS + text	text boxes +	Yes	grounded dialogue, region VQA	RGB-only; 7B-class cost	primary RS-VLM candidate
RS-LLaVA [26]	RGB RS + text	text	Yes	captioning+VQA, LoRA-tuned	no grounding head	captioning/VQA alternative
RSGPT [24]	RGB RS + text	text	Yes	quality RS captions (RSICap)	limited availability	reference/benchmark
SkyEyeGPT [27]	RGB RS + text	text	Yes	unified RS instruction tuning	RGB-only	VQA alternative
LHRS-Bot [28]	RGB RS + text	text	Yes	VGI-enhanced alignment	heavier pipeline	research comparison
EarthGPT [29]	optical/SAR/IR + text	text	Yes	multi-sensor scope	availability/cost	cross-modal VLM candidate
Qwen2.5-VL-class open VLMs	RGB + text	text boxes +	No	strong grounding, active ecosystem	RS domain shift	LoRA-adaptation base [Experimental]
Grounding DINO [15]	RGB + text	boxes	No	open-vocabulary detection	not RS-tuned	zero-shot detection fallback
SAM / SAM 2 [14]	RGB + prompts	masks	No	promptable segmentation	no semantics; RGB	mask refinement from boxes/points
SatMAE [55]	multispectral/features	features	Yes	temporal+multispectral pretraining	encoder only	encoder for fine-tuned heads
SpectralGPT [57]	hyperspectral	features	Yes	spectral foundation model	heavy; HS-focused	[Future] hyperspectral support
Prithvi [58]	HLS multi-spectral	features	Yes	NASA/IBM EO pretraining	encoder only	encoder candidate for segmentation/CD
CROMA [59]	optical+SAR	features	Yes	cross-modal radar-optical learning	encoder only	optical-SAR encoder candidate
DOFA [60]	any-sensor EO	features	Yes	sensor-flexible representation	maturity	[Experimental] unified encoder
SkySense [61]	multimodal EO	features	Yes	large-scale multi-modal pretraining	availability	research reference
ChangeMamba [42]	bi-temporal optical	change masks	Yes	SOTA-class with SSM efficiency	CD training/fine-tune	change-detection specialist

10.3 Selection rationale

For the SIH build: one RS-VLM for perception tasks (GeoChat-class; EarthGPT-class if cross-modal text reasoning is demonstrated), LoRA/QLoRA-adapted on RS instruction data (VRSBench/RSVQA-style) if time permits [Experimental]; specialist, fine-tunable models for segmentation (SegFormer-class) and change detection (BIT/ChangeFormer-class, ChangeMamba as stretch); RemoteCLIP-class embeddings for retrieval/zero-shot support. Full fine-tuning and foundation-model pretraining are explicitly rejected on compute grounds (Section 23).

11 Remote-Sensing Image–Text Datasets

Table 5 catalogues the datasets that matter for training, evaluation, demonstration, and benchmarking. Licensing must be checked per dataset before redistribution or commercial use; most cited datasets are research-licensed and suitable for training and benchmarking, while Sentinel/Landsat data are open [Fact].

Table 5: Key datasets: Task | Sensor | Resolution | Size | Labels | Modality | Main use | SatQuery AI relevance.

Dataset	Task	Sensor	Resolution	Size / labels	Modality	Main use	SatQuery AI relevance
BigEarthNet [53]	multi-label classification	S-1, S-2	10–60 m	~590k patches; CORINE labels	optical+SAR	pretraining, multi-label	domain adaptation; SAR-optical pairing
SEN12MS [52]	classification fusion	S-1, S-2	10 m	~180k scenes; IGBP	optical+SAR	pretraining research	cross-modal experiments
EuroSAT [54]	scene classification	S-2	10 m	27k images; 10 classes	optical	teaching, baselines	quick demo, sanity checks
UCM/Sydney/RSCD/NWPU captions	captioning	S-1, S-2, S-2A, S-2B, S-2C, S-2D, S-2E, S-2F, S-2G, S-2H, S-2I, S-2J, S-2K, S-2L, S-2M, S-2N, S-2O, S-2P, S-2Q, S-2R, S-2S, S-2T, S-2U, S-2V, S-2W, S-2X, S-2Y, S-2Z	10–30 m	0.5k–30k images; 5 captions each	optical	captioning train/eval	captioning fine-tune/eval
RSICap (RS-GPT) [24]	captioning	VHR	various	2.5k human captions	optical	quality captioning	caption quality reference
RSVQA LR/HR [21]	VQA	S-2, aerial	10 m–15 cm	~77k / ~1M QA	optical	VQA train/eval	VQA fine-tune/eval
HRVQA [22]	VQA	aerial	15–150 cm	~1M QA	optical	VHR VQA	VQA evaluation
VRSBench [23]	caption+growth	VHR	~0.5–1 m	~29k images, human-verified	optical	unified LVLM eval	primary VLM benchmark

Dataset	Task	Sensor	Resolution	Size / labels	Modality	Main use	SatQuery AI relevance
RS5M [31]	image–text pretraining	mixed	various	5M pairs	optical	contrastive pretraining	embedding pretraining data
LEVIR-CD [37]	binary change	VHR	0.5 m	637 pairs; building masks	optical	CD train/eval	CD specialist training
WHU-CD [43]	building change	aerial	0.075–0.3 m	1–2 scenes (tiled)	optical	CD eval	CD generalisation check
OSCD [36]	binary change	S-2	10 m	24 city pairs	optical	CD at S-2 scale	demo-scale CD
S2Looking [44]	building change	VHR side-looking	0.5–0.8 m	5k pairs	optical	off-nadir CD	robustness reference
SYSU-CD [45]	change	VHR	0.5 m	20k pairs	optical	CD eval	CD evaluation
SECOND [46]	semantic change	aerial	0.5–3 m	4.7k pairs; 6 classes	optical	semantic CD	change classification
DynamicEarthNet [76]	semantic change time series	Planet	3 m	24 AOIs × 24 months	optical	temporal segmentation	[Future] monitoring
LEVIR-CC [47]	change captioning	VHR	0.5 m	10k pairs; 5 captions	optical	change description	description fine-tune
CDVQA [48]	change VQA	aerial (SECOND)	0.5–3 m	2,968 pairs; >122k QA	optical	change QA	change-VQA benchmark
QAG-360K [49]	change QA + grounding	mixed	0.1–3 m	6,810 pairs; 360k QA-mask triples	optical	grounded change QA	evidence-linked change answers
ChangeChat-87k [50]	change instruction	VHR	0.5 m	87k instructions	optical	temporal instruction tuning	temporal VLM tuning
RSVG [32]	grounding (boxes)	VHR	various	5.5k expressions	optical	REC training	grounding specialist
DIOR-RSVG [33]	grounding (boxes)	VHR (DIOR)	0.5–30 m	27k expressions	optical	REC training	grounding specialist
RRSIS-D [34]	grounding (masks)	VHR	various	17k expression–mask	optical	RES training	mask grounding [Advanced]

Dataset	Task	Sensor	Resolution	Size / labels	Modality	Main use	SatQuery AI relevance
DOTA [65]	object detection	aerial	various	~2.8k images, 15 classes	optical	detection	detector training/eval
DIOR [66]	object detection	VHR	0.5–30 m	23k images, 20 classes	optical	detection	detector training/eval
xView [67]	object detection	WorldView-3	0.3 m	1M+ instances, 60 classes	optical	detection at scale	[Advanced] detector breadth
SpaceNet [68]	buildings/roads	VHR	0.3–0.5 m	multi-city challenges	optical	infrastructure	road/building extraction
SARDet-100K [69]	SAR detection	SAR	various	116k images	SAR	SAR detection	SAR specialist [Advanced]
LoveDA [70]	land-cover segmentation	VHR	0.3 m	6k images, 7 classes	optical	segmentation/segment shift	segmentation training
DeepGlobe [71]	land cover/roads/buildings	VHR	0.5 m	challenge-scale	optical	segmentation	segmentation training
OpenEarthMap [72]	global segmentation	VHR	0.25–0.6 m	5k images, 8 classes	optical	segmentation	segmentation eval
iSAID [73]	instance segmentation	aerial	various	655k instances	optical	instance masks	[Advanced]
FloodNet [74]	flood segmentation	UAV	1.5 cm	2.3k images	optical	disaster segmentation	disaster demo potential
FLAIR [75]	land cover	aerial + S-2 time	0.2 m	country-scale (France)	optical	segmentation + temporal	temporal reference
SSL4EO-S12 [77]	self-supervised pretraining	S-1, S-2	10–60 m	global, multi-season	optical + SAR	pretraining	encoder pretraining data
Geo-Bench [78]	benchmark suite	mixed	various	6 classification + 6 segmentation	optical	GFM evaluation	evaluation reference
PANGAEA [79]	GFM benchmark	mixed	various	multi-task suite	multimodal	GFM evaluation	evaluation reference

Usage plan [Proposal]. Train/fine-tune: LEVIR-CD (+OSCD for S-2 scale), LoveDA / DeepGlobe / OpenEarthMap for segmentation, RSVQA + VRSBench for VLM adaptation, RSVG

/ DIOR-RSVG for grounding. Evaluate: VRSBench (VLM), CDVQA (change QA), LEVIR-CD/WHU-CD (CD), DOTA/DIOR (detection). Demonstrate: Sentinel-1/2 open data over a chosen AOI, plus ISRO imagery via Bhuvan if licensing permits [Assumption].

Part III

System Design

12 Agentic / Task-Routing Approaches for Remote Sensing

12.1 Why not send every query to an LLM

Routing every query to a single LLM/VLM fails on correctness (Section 1.7), cost (per-token pricing for quantitative work that band arithmetic does for free), latency (GPU-bound generation vs. millisecond index computation), and auditability (no intermediate artefacts). The correct pattern is *language in, computation out*: language models parse intent and render explanations; specialist tools do the measuring.

12.2 The routing pipeline

SatQuery AI routing chain (Proposal)

User Query → Intent Classification → Modality Detection → Data Validation → Specialist Tool Selection → Analysis → Evidence Extraction → VLM/LLM Explanation → Visualization → Audit Trace

- **Query classification / intent detection:** map the query to a typed intent (SCENE_VQA, CAPTION, GROUND, INDEX_QUERY, DETECT, SEGMENT, CHANGE_DETECT, CHANGE_VQA, CROSS_MODAL, REPORT, MONITOR). A lightweight classifier (fine-tuned small model or LLM with structured output) suffices; intents are few and well-separated [Proposal].
- **Modality detection and validation:** from metadata — bands, resolution, CRS, polarisation, acquisition date; validate compatibility (bi-temporal needs 2 co-registrable scenes; NDVI needs NIR+Red; SAR questions need SAR input). Fail fast with an explanation, never silently degrade.
- **Specialist model selection:** consult the model registry (name, version, capabilities, input modalities, supported tasks, latency, inference function) and pick the cheapest tool that satisfies the intent.
- **Execution planning:** for multi-step intents, emit a linear or DAG plan (e.g. registration → change detection → building segmentation → intersection → area).
- **Result validation and confidence:** sanity checks (mask coverage, index ranges, registration residual), plus model-reported confidence; low-confidence results are flagged, not hidden.
- **Provenance:** every step appends to the execution trace (Section 21).

12.3 Deterministic router vs. autonomous agent

Critical analysis — is an “agent” actually required?

The concept notes describe a fully agentic workflow (planner, model router, autonomous investigation). For an SIH system, the safe and demonstrable core is a **deterministic router with constrained planning**: a finite intent set, typed tool schemas, and template plans per intent. Fully autonomous multi-step agents add failure modes (plan drift, tool-call loops, unbounded cost) without adding demo value for the core query families. Recommendation: implement the LangGraph-style state machine [81] with *fixed* routing edges per intent and

a bounded “investigate” macro (a pre-defined expansion sequence), reserving open-ended autonomy for post-SIH work [Proposal; Future for full autonomy]. This preserves the concept’s spirit — the user sees planning, tool selection, and traces — while keeping behaviour predictable under judging.

13 The Complete List of 40 Proposed Features

13.1 Extraction method

The concept notes’ “End State” section enumerates exactly forty capabilities; they are reproduced verbatim (names normalised only for grammar) in Tables 6–7. No features are invented; none are silently removed. Section 13.4 then applies critical analysis: duplicates, infrastructure-vs-feature distinctions, and prioritisation.

13.2 Feature specification table

Table 6: The 40 proposed features — specification (Part 1 of 2). Preprocessing abbreviations: Reg = co-registration, CM = cloud mask, Norm = radiometric normalisation, Tile = windowed tiling, RS = resampling.

#	Feature	User problem solved	Input	Modality	Preprocessing	Model / tool
1	Natural-language satellite analysis	No GIS/ML expertise needed to query imagery	text + image(s)	any	metadata parse	query classifier + router
2	Voice-controlled satellite analysis	Hands-free querying	audio + image(s)	any	ASR	speech-to-text + router
3	Remote-sensing VLM	RS-aware perception and dialogue	image + text	optical (RGB)	tiling, render	RGB RS-VLM (GeoChat-class)
4	VQA	Ask questions about a scene	image question	optical	tiling	RS-VLM
5	Captioning	Instant scene description	image	optical	tiling	RS-VLM
6	Visual grounding	Locate the thing asked about	image phrase	optical	tiling	RS-VLM grounding / Grounding-DINO-class
7	Object detection	Find buildings, roads, vehicles	image	optical (SAR adv.)	Tile, Norm	detector (DOTA/DIOR-trained)
8	Segmentation	Pixel-level land-cover objects	image	optical	Tile, Norm, CM	SegFormer-class
9	Land-cover analysis	What covers this area?	image	optical	Tile, CM	segmentation + class stats

#	Feature	User problem solved	Input	Modality	Preprocessing	Model / tool
10	Multitemporal change detection	What changed between dates?	2 images	optical	Reg, Norm	CM, BIT/ChangeFormer-class
11	Change description	Explain the change in words	2 images	optical	as #10	change results → LLM rendering
12	Change VQA	Ask questions about change	2 images + question	optical	as #10	change-VQA (CDVQA-trained)
13	Optical/SAR analysis	Use cloud-proof structural data	SAR (+optical)	SAR, optical	speckle filter, calib.	SAR tools + per-modality models
14	Optical/SAR fusion	Combine spectral + structural evidence	optical + SAR	both	Reg, Norm	late-fusion reasoning + joint explanation
15	GeoTIFF processing	Real georeferenced data support	GeoTIFF	any	metadata, windows	Rasterio/GDAL
16	Metadata intelligence	Know what the data is without asking	any file	any	header parse	metadata extractor + validator
17	Intelligent image tiling	Handle huge scenes	large raster	any	Tile	window engine + reconstruction
18	Geospatial reasoning	Areas, intersections, CRS-correct geometry	raster/vector	any	reprojection	GeoPandas/Shapely/Rasterio
19	Interactive 2D maps	See results on a map	results	any	tile serving	MapLibre/Leaflet frontend
20	Interactive 3D Earth	Geographic context and presentation	results + AOI	any	terrain/imagery layers	CesiumJS
21	Timeline analysis	Scrub through time	≥ 2 dated images	any	Reg, CM	temporal stack + change series
22	Region-aware AI	Ask about a selected area	region query	any	crop to AOI	router + AOI scoping
23	“Ask this region”	One-click regional investigation	polygon query	any	crop, context build	router + region pipeline
24	Autonomous investigation	Multi-step drill-down from one answer	prior result	any	plan template	bounded investigation macro
25	Agentic model routing	Right tool for each query	query + registry	any	—	deterministic router (LangGraph)

#	Feature	User problem solved	Input	Modality	Preprocessing	Model / tool
26	Specialist model registry	Manage many models cleanly	model zoo	any	—	registry service
27	Evidence graph	Every claim linked to evidence	results	any	—	evidence engine + store
28	Confidence estimation	Know when to trust an answer	model outputs	any	calibration	confidence aggregator
29	Execution traces	Audit how an answer was made	run logs	any	—	trace logger
30	Model observability	Latency, versions, health	telemetry	any	—	metrics/logging stack
31	Research/benchmark mode	Compare models inside the app	datasets + models	any	harness	evaluation framework
32	Voice interaction	Conversational voice control	audio	any	ASR (+TTS)	ASR + router (+TTS)
33	Mission creation	Save investigations as reusable missions	AOI + config	any	—	mission store
34	Continuous monitoring	Watch an AOI over time	AOI + schedule	any	Reg, CM	scheduler + change pipeline
35	Automated alerts	Get notified on significant change	monitoring events	any	thresholds	alerting service
36	Mission history	Revisit past missions	store	any	—	database
37	Investigation history	Revisit past analyses	store	any	—	database
38	Professional intelligence reports	Share findings formally	investigation	any	—	report generator (PDF)
39	PDF/JSON/GeoJSON export	Export results elsewhere	results	any	—	export services
40	Persistent AI workspace	Keep data, chats, evidence across sessions	all	any	—	database + object storage

13.3 Engineering assessment table

Categories: **MVP** = core minimum viable product; **SIH** = strong SIH-demo feature; **ADV** = advanced (build if time permits); **RES** = research-grade; **FUT** = future extension. Difficulty: L/M/H/VH. Reliability: expected output trustworthiness when built as specified.

Table 7: The 40 proposed features — engineering assessment (Part 2 of 2). Output/visualisation and priority.

#	Output → visualisation	Dependencies	Diff.	Reliability	Demo value	Research value	Category
1	answer text → chat panel	15,16,25	M	high (routed)	very high	medium	MVP
2	text → chat	1 + ASR	M	medium (ASR errors)	medium	low	FUT
3	RS-aware answers → chat	15,17	H	medium–high	high	high	MVP
4	answer → chat	3	M	medium–high	very high	high	MVP
5	caption → chat	3	M	medium–high	high	medium	MVP
6	boxes/masks → map overlay	3,19	H	medium	very high	high	SIH
7	boxes → map overlay	15,17,19	H	medium–high	high	medium	SIH
8	masks → map overlay	15,17,19	H	medium–high	high	medium	SIH
9	class map + stats → map + chart	8	M	high	high	low	MVP
10	change mask → change map	15,Reg	H	medium–high	very high	high	SIH
11	paragraph → chat/report	10,11-renderer	M	high (if grounded)	high	medium	SIH
12	answer → chat	10 (+CDVQA model)	H	medium	high	high	ADV
13	SAR products → side-by-side	15, SAR prep	H	medium	high	high	SIH
14	joint conclusion → dual evidence	13, Reg	VH	medium	very high	high	ADV
15	parsed raster → viewer	—	M	high	medium	low	MVP
16	metadata card → UI panel	15	L	high	medium	low	MVP
17	tiles → seamless result	15	M	high	low	low	MVP
18	areas/geometry → map, stats	15	M	high	medium	low	MVP
19	interactive map → UI	18	M	high	very high	low	MVP
20	3D globe → UI	19	H	high	high	low	ADV

#	Output → visualisation	Dependencies	Diff.	Reliability	Demo value	Research value	Category
21	time scrubber → UI	10,19	M	medium–high	high	medium	ADV
22	scoped answers → map+chat	1,19	M	medium–high	high	medium	SIH
23	region report → map+chat	22	M	medium–high	very high	medium	SIH
24	expanded analysis → trace UI	25,27	H	medium	high	medium	ADV
25	routed plan → trace UI	26	M	high	high	medium	MVP
26	registry API → observatory page	—	L	high	medium	low	MVP
27	evidence links → evidence page	29	M	high	high	high	SIH
28	confidence badge → UI	per-model scores	M	medium	high	medium	SIH
29	trace view → UI	25	L	high	high	medium	MVP
30	dashboards → observatory	26,29	M	high	medium	low	ADV
31	benchmark tables → UI	26, datasets	H	high	low	high	FUT
32	voice I/O → UI	2	M	medium	medium	low	FUT
33	saved mission → UI	40	M	high	medium	low	ADV
34	monitoring runs → timeline	10,33,40	VH	medium	medium	medium	FUT
35	notifications → UI	34	M	medium	medium	low	FUT
36	mission list → UI	33,40	L	high	low	low	ADV
37	history list → UI	40	L	high	medium	low	ADV
38	PDF report → download	27,29	M	high	very high	low	SIH
39	files → download	18	L	high	medium	low	MVP
40	persistent sessions → UI	DB + storage	M	high	medium	low	SIH

13.4 Critical analysis of the feature list

Duplicates, overlaps, and category corrections

1. **Exact duplicate:** #2 (voice-controlled analysis) and #32 (voice interaction) are the same capability; they are merged into one roadmap item (FUT).
2. **Near-duplicates:** #22 (region-aware AI) and #23 (“ask this region”) are one mechanism at two granularities — implemented once, exposed twice. #36, #37, #40 (mission history, investigation history, persistent workspace) are three views of one persistence layer.
3. **Infrastructure mislabelled as features:** #15 (GeoTIFF), #16 (metadata), #17 (tiling), #18 (geospatial reasoning), #26 (registry), #29 (traces) are enabling infrastructure — mandatory, but they are *modules*, not user-facing features. The taxonomy (Section 14) reflects this.
4. **Missing from the concept notes:** a spectral-index engine (NDVI/NDWI/NDBI/NBR) — required by the project’s own example queries; cloud masking; CRS/reprojection services; and a data-ingestion story beyond upload (STAC/Bhuvan) [Proposal: added as modules, Section 16].
5. **Unrealistic for SIH as stated:** #34 continuous monitoring (needs data acquisition automation + scheduling + storage budget), #35 alerts, #31 benchmark mode, #24 full autonomy, #20 3D Earth (high effort, low analytical value), #2/#32 voice. These are staged ADV/FUT.

13.5 Feature dependency graph

Figure 1 shows the build dependencies among feature clusters.

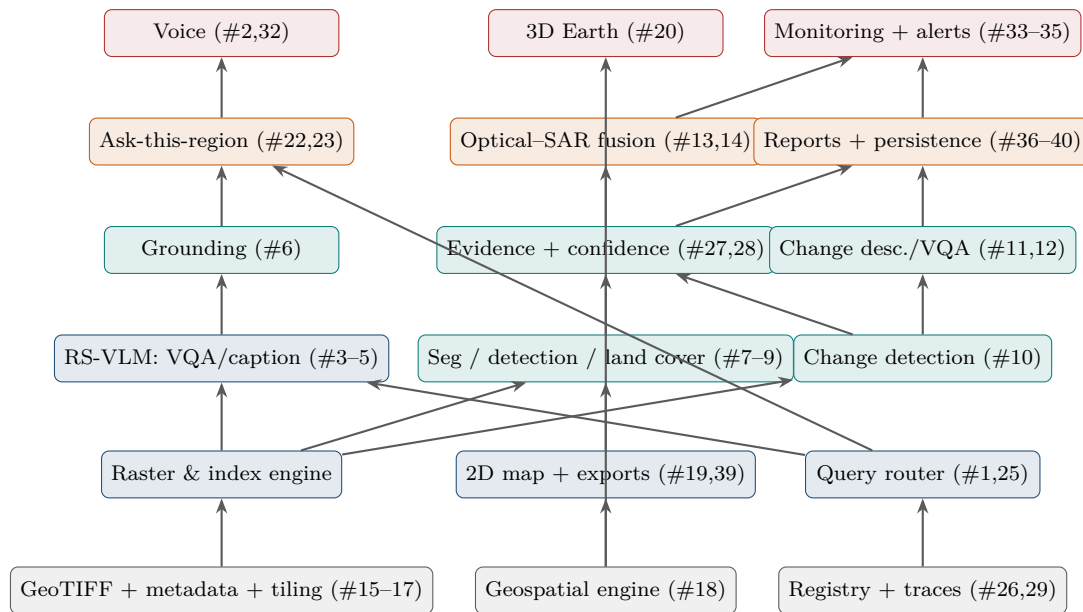


Figure 1: Feature dependency graph (build bottom-up). Colour: grey = infrastructure, blue = MVP, teal = SIH-demo, orange = advanced, red = future.

14 Feature Taxonomy

The forty features group into nine logical domains (Table 8); categories follow what the concept notes actually support, with the index engine added as the one justified extension.

Table 8: Feature taxonomy by domain.

Domain	Features (#)
Image understanding & VQA	RS-VLM (3), VQA (4), captioning (5)
Text-to-region grounding	Visual grounding (6), region-aware AI (22), ask-this-region (23)
Detection & segmentation	Object detection (7), segmentation (8), land-cover analysis (9)
Spectral analysis	Index engine (added; serves NDVI/NDWI/NDBI/NBR queries)
Temporal analysis & change	Change detection (10), change description (11), change VQA (12), timeline (21)
SAR & cross-modal	Optical/SAR analysis (13), fusion (14)
Geospatial & data infrastructure	GeoTIFF (15), metadata (16), tiling (17), geospatial reasoning (18), exports (39)
Agentic routing & explainability	NL analysis (1), routing (25), registry (26), evidence graph (27), confidence (28), traces (29), observability (30), autonomous investigation (24), benchmark mode (31)
Interaction, reporting & persistence	2D maps (19), 3D Earth (20), voice (2, 32), missions (33), monitoring (34), alerts (35), histories (36, 37), reports (38), workspace (40)

15 Remote-Sensing Analysis Pipeline

15.1 The canonical 20-stage pipeline

- S1. Data ingestion** — upload or reference (STAC/Bhuvan [Future]); store raw file immutably.
- S2. File identification** — format sniffing (GeoTIFF/JP2/PNG), band count, driver.
- S3. Metadata extraction** — sensor hints, acquisition date, processing level, nodata, scale/offset.
- S4. CRS detection** — EPSG lookup; reject or repair missing CRS (with user confirmation).
- S5. Band identification** — map band indices to wavelengths (Sentinel-2 templates; user hint fallback).
- S6. Quality checking** — nodata fraction, histogram sanity, resolution report.
- S7. Cloud / no-data handling** — SCL/s2cloudless masking for optical; flag coverage.
- S8. Spatial normalisation** — reproject to analysis CRS; align grids.
- S9. Co-registration** — for pairs: geometric alignment + residual estimate; abort/warn above threshold.
- S10. Resampling** — harmonise band resolutions (e.g. SWIR 20 m $\rightarrow$ 10 m grid).
- S11. Tiling** — windowed reads (e.g. 512×512 , 10% overlap).
- S12. Feature extraction** — indices, embeddings, per-tile model features.
- S13. Specialist analysis** — detection / segmentation / change / SAR tools.
- S14. VLM reasoning** — VQA/captioning/grounding over rendered tiles or regions.
- S15. Evidence localisation** — bind outputs to georeferenced regions (masks, boxes).
- S16. Answer generation** — structured results $\rightarrow$ constrained LLM rendering.
- S17. Visualisation** — overlays, change maps, charts, side-by-side views.
- S18. Confidence estimation** — per-tool scores + validation checks $\rightarrow$ aggregate badge.
- S19. Provenance logging** — append full trace (inputs, versions, parameters, timings).
- S20. Final response** — answer + evidence + confidence + trace + export handles.

15.2 Pipeline per input configuration

Table 9: Stage activation per input type (● = active, ○ = conditional).

Stages	Optical	SAR	Opt+SAR	Bi-temporal	Text+img	Text+2img
S1–S6 ingestion/metadata	●	●	●	●	●	●
S7 cloud handling	●	—	○	●	●	●
S8–S10 registration/resample	○	○	●	●	○	●
S11 tiling	●	●	●	●	●	●
S12 indices/features	●	○	●	●	○	○
S13 specialist analysis	○	●	●	●	○	●
S14 VLM reasoning	○	○	●	○	●	●
S15–S20 evidence→response	●	●	●	●	●	●

SAR-specific stages replace S7 with radiometric calibration, speckle filtering (e.g. refined Lee), and terrain-flattening awareness; optical–SAR runs both branches and joins at S15 (evidence localisation) per the late-fusion policy (Section 9).

16 System Modules

Table 10: Software modules: purpose, I/O, technology, owner role, complexity.

#	Module	Purpose / I/O	Technologies	Owner role	Compl.	Depends on
M1	Frontend / UI	chat, upload, map, evidence, report views; in: user events; out: API calls	Next.js, React, TypeScript, Tailwind, Zustand, TanStack Query	Frontend	H	M18
M2	Upload & ingestion	file intake, validation, storage; in: files; out: asset records	FastAPI, object storage	Backend	M	—
M3	Geospatial preprocessing	CRS, reprojection, registration, resampling; in: rasters; out: aligned rasters	Rasterio, GDAL, pyproj	Geospatial	H	M2
M4	Raster engine	windowed reads, tiling, reconstruction; in: rasters; out: tiles/products	Rasterio, xarray, NumPy	Geospatial	M	M3

#	Module	Purpose / I/O	Technologies	Owner	role	Compl.	Depends on
M5	Spectral-index engine	NDVI/EVI/SAVI/NDWI/NDVI ₂ /NDVI ₃ /NDVI ₄ /NDVI ₅ /NDVI ₆ /NDVI ₇ /NDVI ₈ /NDVI ₉ /NDVI ₁₀ /NDVI ₁₁ /NDVI ₁₂ /NDVI ₁₃ /NDVI ₁₄ /NDVI ₁₅ /NDVI ₁₆ /NDVI ₁₇ /NDVI ₁₈ /NDVI ₁₉ /NDVI ₂₀ /NDVI ₂₁ /NDVI ₂₂ /NDVI ₂₃ /NDVI ₂₄ /NDVI ₂₅ /NDVI ₂₆ /NDVI ₂₇ /NDVI ₂₈ /NDVI ₂₉ /NDVI ₃₀ /NDVI ₃₁ /NDVI ₃₂ /NDVI ₃₃ /NDVI ₃₄ /NDVI ₃₅ /NDVI ₃₆ /NDVI ₃₇ /NDVI ₃₈ /NDVI ₃₉ /NDVI ₄₀ /NDVI ₄₁ /NDVI ₄₂ /NDVI ₄₃ /NDVI ₄₄ /NDVI ₄₅ /NDVI ₄₆ /NDVI ₄₇ /NDVI ₄₈ /NDVI ₄₉ /NDVI ₅₀ /NDVI ₅₁ /NDVI ₅₂ /NDVI ₅₃ /NDVI ₅₄ /NDVI ₅₅ /NDVI ₅₆ /NDVI ₅₇ /NDVI ₅₈ /NDVI ₅₉ /NDVI ₆₀ /NDVI ₆₁ /NDVI ₆₂ /NDVI ₆₃ /NDVI ₆₄ /NDVI ₆₅ /NDVI ₆₆ /NDVI ₆₇ /NDVI ₆₈ /NDVI ₆₉ /NDVI ₇₀ /NDVI ₇₁ /NDVI ₇₂ /NDVI ₇₃ /NDVI ₇₄ /NDVI ₇₅ /NDVI ₇₆ /NDVI ₇₇ /NDVI ₇₈ /NDVI ₇₉ /NDVI ₈₀ /NDVI ₈₁ /NDVI ₈₂ /NDVI ₈₃ /NDVI ₈₄ /NDVI ₈₅ /NDVI ₈₆ /NDVI ₈₇ /NDVI ₈₈ /NDVI ₈₉ /NDVI ₉₀ /NDVI ₉₁ /NDVI ₉₂ /NDVI ₉₃ /NDVI ₉₄ /NDVI ₉₅ /NDVI ₉₆ /NDVI ₉₇ /NDVI ₉₈ /NDVI ₉₉ /NDVI ₁₀₀ /NDVI ₁₀₁ /NDVI ₁₀₂ /NDVI ₁₀₃ /NDVI ₁₀₄ /NDVI ₁₀₅ /NDVI ₁₀₆ /NDVI ₁₀₇ /NDVI ₁₀₈ /NDVI ₁₀₉ /NDVI ₁₁₀ /NDVI ₁₁₁ /NDVI ₁₁₂ /NDVI ₁₁₃ /NDVI ₁₁₄ /NDVI ₁₁₅ /NDVI ₁₁₆ /NDVI ₁₁₇ /NDVI ₁₁₈ /NDVI ₁₁₉ /NDVI ₁₂₀ /NDVI ₁₂₁ /NDVI ₁₂₂ /NDVI ₁₂₃ /NDVI ₁₂₄ /NDVI ₁₂₅ /NDVI ₁₂₆ /NDVI ₁₂₇ /NDVI ₁₂₈ /NDVI ₁₂₉ /NDVI ₁₃₀ /NDVI ₁₃₁ /NDVI ₁₃₂ /NDVI ₁₃₃ /NDVI ₁₃₄ /NDVI ₁₃₅ /NDVI ₁₃₆ /NDVI ₁₃₇ /NDVI ₁₃₈ /NDVI ₁₃₉ /NDVI ₁₄₀ /NDVI ₁₄₁ /NDVI ₁₄₂ /NDVI ₁₄₃ /NDVI ₁₄₄ /NDVI ₁₄₅ /NDVI ₁₄₆ /NDVI ₁₄₇ /NDVI ₁₄₈ /NDVI ₁₄₉ /NDVI ₁₅₀ /NDVI ₁₅₁ /NDVI ₁₅₂ /NDVI ₁₅₃ /NDVI ₁₅₄ /NDVI ₁₅₅ /NDVI ₁₅₆ /NDVI ₁₅₇ /NDVI ₁₅₈ /NDVI ₁₅₉ /NDVI ₁₆₀ /NDVI ₁₆₁ /NDVI ₁₆₂ /NDVI ₁₆₃ /NDVI ₁₆₄ /NDVI ₁₆₅ /NDVI ₁₆₆ /NDVI ₁₆₇ /NDVI ₁₆₈ /NDVI ₁₆₉ /NDVI ₁₇₀ /NDVI ₁₇₁ /NDVI ₁₇₂ /NDVI ₁₇₃ /NDVI ₁₇₄ /NDVI ₁₇₅ /NDVI ₁₇₆ /NDVI ₁₇₇ /NDVI ₁₇₈ /NDVI ₁₇₉ /NDVI ₁₈₀ /NDVI ₁₈₁ /NDVI ₁₈₂ /NDVI ₁₈₃ /NDVI ₁₈₄ /NDVI ₁₈₅ /NDVI ₁₈₆ /NDVI ₁₈₇ /NDVI ₁₈₈ /NDVI ₁₈₉ /NDVI ₁₉₀ /NDVI ₁₉₁ /NDVI ₁₉₂ /NDVI ₁₉₃ /NDVI ₁₉₄ /NDVI ₁₉₅ /NDVI ₁₉₆ /NDVI ₁₉₇ /NDVI ₁₉₈ /NDVI ₁₉₉ /NDVI ₂₀₀ /NDVI ₂₀₁ /NDVI ₂₀₂ /NDVI ₂₀₃ /NDVI ₂₀₄ /NDVI ₂₀₅ /NDVI ₂₀₆ /NDVI ₂₀₇ /NDVI ₂₀₈ /NDVI ₂₀₉ /NDVI ₂₁₀ /NDVI ₂₁₁ /NDVI ₂₁₂ /NDVI ₂₁₃ /NDVI ₂₁₄ /NDVI ₂₁₅ /NDVI ₂₁₆ /NDVI ₂₁₇ /NDVI ₂₁₈ /NDVI ₂₁₉ /NDVI ₂₂₀ /NDVI ₂₂₁ /NDVI ₂₂₂ /NDVI ₂₂₃ /NDVI ₂₂₄ /NDVI ₂₂₅ /NDVI ₂₂₆ /NDVI ₂₂₇ /NDVI ₂₂₈ /NDVI ₂₂₉ /NDVI ₂₃₀ /NDVI ₂₃₁ /NDVI ₂₃₂ /NDVI ₂₃₃ /NDVI ₂₃₄ /NDVI ₂₃₅ /NDVI ₂₃₆ /NDVI ₂₃₇ /NDVI ₂₃₈ /NDVI ₂₃₉ /NDVI ₂₄₀ /NDVI ₂₄₁ /NDVI ₂₄₂ /NDVI ₂₄₃ /NDVI ₂₄₄ /NDVI ₂₄₅ /NDVI ₂₄₆ /NDVI ₂₄₇ /NDVI ₂₄₈ /NDVI ₂₄₉ /NDVI ₂₅₀ /NDVI ₂₅₁ /NDVI ₂₅₂ /NDVI ₂₅₃ /NDVI ₂₅₄ /NDVI ₂₅₅ /NDVI ₂₅₆ /NDVI ₂₅₇ /NDVI ₂₅₈ /NDVI ₂₅₉ /NDVI ₂₆₀ /NDVI ₂₆₁ /NDVI ₂₆₂ /NDVI ₂₆₃ /NDVI ₂₆₄ /NDVI ₂₆₅ /NDVI ₂₆₆ /NDVI ₂₆₇ /NDVI ₂₆₈ /NDVI ₂₆₉ /NDVI ₂₇₀ /NDVI ₂₇₁ /NDVI ₂₇₂ /NDVI ₂₇₃ /NDVI ₂₇₄ /NDVI ₂₇₅ /NDVI ₂₇₆ /NDVI ₂₇₇ /NDVI ₂₇₈ /NDVI ₂₇₉ /NDVI ₂₈₀ /NDVI ₂₈₁ /NDVI ₂₈₂ /NDVI ₂₈₃ /NDVI ₂₈₄ /NDVI ₂₈₅ /NDVI ₂₈₆ /NDVI ₂₈₇ /NDVI ₂₈₈ /NDVI ₂₈₉ /NDVI ₂₉₀ /NDVI ₂₉₁ /NDVI ₂₉₂ /NDVI ₂₉₃ /NDVI ₂₉₄ /NDVI ₂₉₅ /NDVI ₂₉₆ /NDVI ₂₉₇ /NDVI ₂₉₈ /NDVI ₂₉₉ /NDVI ₃₀₀ /NDVI ₃₀₁ /NDVI ₃₀₂ /NDVI ₃₀₃ /NDVI ₃₀₄ /NDVI ₃₀₅ /NDVI ₃₀₆ /NDVI ₃₀₇ /NDVI ₃₀₈ /NDVI ₃₀₉ /NDVI ₃₁₀ /NDVI ₃₁₁ /NDVI ₃₁₂ /NDVI ₃₁₃ /NDVI ₃₁₄ /NDVI ₃₁₅ /NDVI ₃₁₆ /NDVI ₃₁₇ /NDVI ₃₁₈ /NDVI ₃₁₉ /NDVI ₃₂₀ /NDVI ₃₂₁ /NDVI ₃₂₂ /NDVI ₃₂₃ /NDVI ₃₂₄ /NDVI ₃₂₅ /NDVI ₃₂₆ /NDVI ₃₂₇ /NDVI ₃₂₈ /NDVI ₃₂₉ /NDVI ₃₃₀ /NDVI ₃₃₁ /NDVI ₃₃₂ /NDVI ₃₃₃ /NDVI ₃₃₄ /NDVI ₃₃₅ /NDVI ₃₃₆ /NDVI ₃₃₇ /NDVI ₃₃₈ /NDVI ₃₃₉ /NDVI ₃₄₀ /NDVI ₃₄₁ /NDVI ₃₄₂ /NDVI ₃₄₃ /NDVI ₃₄₄ /NDVI ₃₄₅ /NDVI ₃₄₆ /NDVI ₃₄₇ /NDVI ₃₄₈ /NDVI ₃₄₉ /NDVI ₃₅₀ /NDVI ₃₅₁ /NDVI ₃₅₂ /NDVI ₃₅₃ /NDVI ₃₅₄ /NDVI ₃₅₅ /NDVI ₃₅₆ /NDVI ₃₅₇ /NDVI ₃₅₈ /NDVI ₃₅₉ /NDVI ₃₆₀ /NDVI ₃₆₁ /NDVI ₃₆₂ /NDVI ₃₆₃ /NDVI ₃₆₄ /NDVI ₃₆₅ /NDVI ₃₆₆ /NDVI ₃₆₇ /NDVI ₃₆₈ /NDVI ₃₆₉ /NDVI ₃₇₀ /NDVI ₃₇₁ /NDVI ₃₇₂ /NDVI ₃₇₃ /NDVI					

#	Module	Purpose / I/O	Technologies	Owner role	Compl.	Depends on
M17	Evidence / provenance	claim-region-model links, traces; in: all outputs; out: evidence graph	PostgreSQL/PostGIS, JSON-LD-ish records	Backend	M	M16
M18	Visualisation / map engine	tiles, overlays, change maps; in: products; out: web layers	MapLibre/Leaflet, titiler-style tiles, (CesiumJS ADV)	Frontend	H	M4
M19	Backend API	REST + Web-Socket, sessions, auth-lite	FastAPI, Web-Sockets	Backend	M	all
M20	Database	assets, missions, history, evidence	PostgreSQL/PostGIS	Backend	M	—
M21	Model serving	registry, versioning, GPU workers, CPU fallback	HF, ONNX Runtime, Docker	MLOps	H	M26-registry
M22	Evaluation framework	metrics harness, benchmark runs	TorchGeo-style eval, scripts	CV/VLM	M	M21

Each module exposes a typed interface (JSON schema in / artefacts out), which is what makes the router testable and the traces meaningful [Proposal].

17 What Each Team Member Needs to Study

The concept notes specify roles, not named members; assignments below are role-based (no personal information invented). Prerequisite chains follow the learning roadmap (Section 27).

Table 11: Team learning matrix: Role → Topics → Libraries → Models → Practice → Owns → Dependencies.

Role	Topics	Libraries	Models	Practice project	Owns	Depends on
Remote sensing / geospatial	EM 4 spectrum, resolutions, SAR basics, CRS, indices, co-registration	Rasterio, GDAL, GeoPandas, Shapely, pyproj, xarray	— (index math)	Build an NDVI/NDWI + registration notebook on S-1/S-2	M3–M5, S7–S10 stages	Python, NumPy
Computer vision	CNNs, ViTs, detection, segmentation, change detection, registration	PyTorch, torchvision, OpenCV, Albumentations	U-Net, SegFormer, BIT, ChangeFormer, (ChangeMamba)	Train + evaluate a LEVIR-CD change model	M6–M9	RS/geospatial role
VLM GenAI	/ transformers, CLIP, LLaVA architecture, instruction tuning, LoRA/QLoRA, grounding	HF Transformers, PEFT, Datasets	CLIP, LLaVA, GeoChat, RS-LLaVA, Grounding DINO, SAM	LoRA-adapt an RS-VLM on VRSBench subset	M10–M13	CV role
Backend API	/ async Python, REST, WebSockets, storage, queues	FastAPI, PostgreSQL/PostGIS, Redis (opt.), Docker	—	Upload → process → read API with trace records	M1, M17, M19, M20	
Frontend / visualisation	React, maps, raster overlays, streaming UI	Next.js, TypeScript, Tailwind, MapLibre, (CesiumJS)	—	Interactive map with mask/change overlays + chat	M1, M18	Backend API
Data engineering	datasets, formats, splits, licensing, STAC	GDAL, STAC tools, HF Datasets	—	Dataset cards + loaders for LEVIR-CD, VRSBench, SEN12MS	dataset pipeline	RS role
MLOps / deployment	GPU inference, quantisation, serving, monitoring	Docker, ONNX Runtime, vLLM (opt.), MLflow (opt.)	all served models	One-command GPU inference service with fallbacks	M21	CV+VLM roles
Integration / agentic AI	tool calling, structured outputs, state machines, traces, evaluation of routing	LangGraph, Pydantic, LLM APIs	router classifier	Deterministic router over 6 intents with full traces	M14–M16	Backend, VLM

For a typical 6-person SIH team, merge: Data engineering → RS/geospatial; MLOps → Backend. The two roles that cannot be merged without risk are *geospatial* and *VLM* — they share almost no prerequisite overlap.

18 Technical Architecture

18.1 Overview

Architecture in one paragraph (Proposal)

A Next.js frontend talks to a FastAPI backend over REST (uploads, CRUD) and WebSockets (streaming analysis progress). The backend hosts the LangGraph router, which dispatches to specialist services (raster/index engine, CV models, VLM serving) running as Python workers

with GPU access. Rasters live in object storage; structured state (assets, missions, evidence, traces) lives in PostgreSQL/PostGIS. Every run writes a provenance record that the frontend renders as the execution trace and evidence views.

18.2 Frontend

Next.js + React + TypeScript + Tailwind [Proposal]. Map: **MapLibre GL** (2D raster overlays, boxes, masks, change maps) as default; Leaflet acceptable; **CesiumJS** only for the ADV 3D Earth. Charts for index/area statistics. Key inclusion rationale: MapLibre handles georeferenced overlays natively; Cesium's 3D adds no analytical capability — hence ADV, not MVP.

18.3 Backend

FastAPI (async Python) — chosen because the entire geospatial/ML stack is Python; one language end-to-end removes an integration layer. REST for uploads/CRUD; **WebSockets** for streaming execution-trace events and progress (judges see the plan unfold live — high demo value). Background jobs via a simple task queue; heavy inference on GPU workers.

18.4 AI layer

Strictly separated concerns: **VLM** (perception + explanation rendering), **LLM-class classifier** (intent), **specialist CV models** (detection, segmentation, change), **embeddings** (RemoteCLIP-class for retrieval/zero-shot support), **tool execution** (deterministic Python tools: indices, area stats, registration). No RAG document store in MVP — the knowledge the system needs is in the imagery and tool outputs, not in documents [Proposal].

18.5 Geospatial layer

Rasterio (Pythonic raster I/O — primary), **GDAL** (underneath Rasterio; CLI for conversions), **GeoPandas** + **Shapely** (vectors, intersections, area), **pyproj** (CRS), **xarray** (multi-band/time stacks, optional), **PostGIS** (spatial queries over AOIs/detections), **STAC** (deferred to FUT for catalogue ingestion). TorchGeo [80] recommended for dataset loaders and samplers.

18.6 Storage

PostgreSQL/PostGIS (assets, AOIs, detections, missions, evidence, traces) + object storage or filesystem bucket (raw rasters, products, report PDFs) + in-memory/Redis cache for rendered tiles (optional). No vector database in MVP (no semantic-search requirement).

18.7 Model serving

Hugging Face Transformers for VLM + PEFT/LoRA weights; ONNX Runtime as CPU fallback for specialist models; vLLM/TorchServe unnecessary at SIH scale. Quantisation (8-bit/4-bit via QLoRA-style loading) to fit a 7B-class VLM on a single ≤ 16 GB GPU; specialist models are small enough to co-reside. CPU fallback exists for everything except acceptable-latency VLM chat — which is why VLM calls are isolated behind M13 with graceful degradation [Proposal].

18.8 Explicitly excluded (with reasons)

Node.js backend (splits the language stack); Kubernetes (SIH overkill — Docker Compose suffices); full microservice mesh (modules are Python packages behind one API first; split only if GPU contention demands); vector database (no RAG); STAC ingestion (FUT); 3D Earth (ADV); voice (FUT).

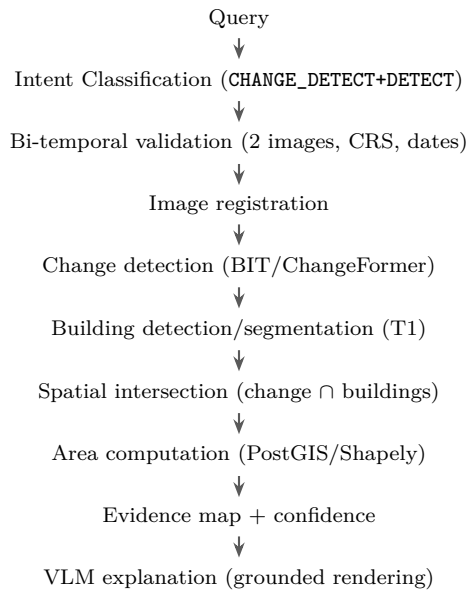
19 Query Understanding and Task Routing

19.1 Query interpretation

The interpreter extracts: intent, task type, target objects, spatial region, temporal range, required modality, image count, and requested output. Example (from the concept notes): “*Has the built-up area increased between these two images?*” → **task:** CHANGE_VQA; **images:** 2; **temporal:** T0→T1; **target:** built-up; **output:** answer + change map + evidence.

19.2 Routing examples

Example 1 — bi-temporal construction query. “*Compare these two images and show where new buildings appeared.*”



Example 2 — vegetation health. “*Show me areas with unhealthy vegetation.*” → intent INDEX_QUERY → band validation (NIR+Red present? L2A?) → cloud mask → NDVI → threshold + connected components → stressed-region mask + stats → evidence → explanation. No VLM call until the final rendering.

Example 3 — water localisation. “*Where are the water bodies?*” → GROUND or INDEX_QUERY: if the scene is urban, prefer MNDWI (built-up suppression) and/or water segmentation; return mask + area; optionally cross-check with VLM grounding for the phrase “water body”.

Example 4 — cross-modal. “*What changed between the optical and SAR images?*” → CROSS_MODAL → modality validation → co-registration → per-sensor analysis (optical indices/land cover; SAR backscatter/structure) → joint interpretation → dual-evidence explanation. The answer must say *which sensor supports which claim*.

Example 5 — pure perception. “*Is there an airport in this image?*” → SCENE_VQA → tiling → RS-VLM → answer + (if affirmative) grounding follow-up offer.

Example 6 — evidence recall. “*Show the region responsible for your answer.*” → EVIDENCE_RECALL → evidence store lookup → map highlight. No model inference at all — proof that the evidence layer is real, not decorative.

20 Evidence-Grounded Answers

Design principle: the LLM explains evidence; it never invents it

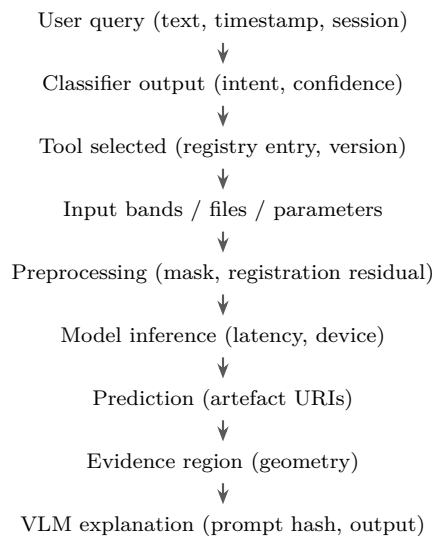
The answer pipeline is strictly: **specialist model** → **structured result** → **spatial/temporal evidence** → **validation** → **LLM explanation**. The generation step receives only validated structured results (masks, statistics, confidence) and is instructed to verbalise exactly them.

Every answer object contains: answer text; confidence; highlighted region(s) (georeferenced); relevant evidence artefacts (index map, change mask, boxes); model/tool used with version; input imagery identifiers; acquisition timestamps; and the processing trace. Example (concept-note target): *“Built-up area increased by approximately 18.4%; 14.2 hectares of new built-up regions, primarily north-east; confidence 91%”* — where 18.4% and 14.2 ha are *computed* from the georeferenced mask, and 91% aggregates model confidence and validation checks.

Why this reduces hallucination and increases trust [Fact + Proposal]: generation conditioned on structured, validated inputs has a drastically smaller fabrication surface than free-form answering; every numeric claim is traceable to a computation; every spatial claim to pixels. This is the same principle behind grounded-VQA research [49] and retrieval-grounded generation generally. It also enables the “low confidence” UX: when evidence is insufficient, the system says so and offers alternatives (“use another sensor”), which is more valuable than a fluent guess.

21 Explainability and Provenance

21.1 The auditable analysis trace



21.2 Provenance requirements

- **Reproducibility:** given the trace, the run must be re-executable — store model versions, parameters, random seeds, and input hashes.
- **Model/version logging:** registry entries are immutable; a run references `model@version`.
- **Input metadata:** file hashes, CRS, acquisition dates, processing level.
- **Timestamps and latency** per stage (feeds model observability, feature #30).
- **Confidence:** per-stage scores + aggregation rule recorded, not just shown.
- **Intermediate outputs:** masks, index maps, and registration residuals are retained as addressable artefacts (object storage URIs in the trace).
- **Audit trail:** append-only; exports (PDF/JSON) embed the trace ID.

This converts SatQuery AI from a demo into an instrument: an ISRO evaluator can pick any claim in any report and walk it back to pixels, parameters, and model versions [Proposal].

Part IV

Engineering Governance

22 Evaluation Framework

SatQuery AI is evaluated *per layer*, not only on final chat answers — matching the concept notes’ multimodal-evaluation requirement.

22.1 Remote-sensing metrics

Segmentation/change: Intersection-over-Union $\text{IoU} = \frac{|P \cap G|}{|P \cup G|}$, mean IoU, Dice = $\frac{2|P \cap G|}{|P| + |G|}$, precision, recall, $F_1 = \frac{2PR}{P+R}$, pixel accuracy; change detection reports F1/IoU on the change class (LEVIR-CD convention). Detection: mAP at IoU thresholds (DOTA/DIOR convention).

22.2 VLM / VQA metrics

VQA: accuracy / exact match on RSVQA, VRSBench, CDVQA splits. Captioning/description: CIDEr, BLEU, ROUGE, METEOR as proxies *plus* human evaluation on a fixed 50-scene panel (fluency, correctness, terminology). Semantic-similarity scoring for free-form answers.

22.3 Grounding metrics

Box IoU@0.5 accuracy (REC convention), pointing accuracy (point-in-region), mask IoU for RES outputs; evaluated on RSVG/DIOR-RSVG and VRSBench grounding splits.

22.4 System-level metrics

Routing accuracy (intent classification on a hand-labelled query set — target $\geq 95\%$ on the six core intents); tool-selection accuracy; end-to-end latency per query family; GPU memory; throughput; failure rate (crashes, validation aborts); fallback activation rate.

22.5 User-level metrics

Task completion on scripted analyst tasks; analyst time saved vs. manual workflow; interpretability rating (can the user find the evidence for a claim in $<30\text{ s}$?); usability (SUS-style questionnaire).

Why NLP metrics alone are insufficient

BLEU/ROUGE/CIDEr reward phrasing overlap, not correctness. A caption can score well while mislocating every object; a change description can be fluent and numerically wrong. SatQuery AI answers carry *quantitative* and *spatial* commitments, so evaluation must include geospatial metrics (IoU, area error), grounding accuracy, and routing correctness — NLP scores are reported but never sufficient [Fact + Proposal].

23 Design Decisions

Table 12: Design-decision log: Decision → Alternatives → Reason → Trade-off.

Decision	Alternatives	Reason for selection	Trade-off
VLM + specialist tools	VLM-only; tools-only UI	VLM alone hallucinates and cannot quantify; tools alone are unusable by non-experts	more moving parts; needs routing
Deterministic router	autonomous agent (ReAct-style); hard-coded menu	predictable under judging; auditable; matches finite intent set	less flexible to novel queries; “agentic” is bounded
FastAPI (Python)	Node.js/Express; Django	one language across geospatial+ML+API; async + WebSocket built-in	Django admin lost; Node ecosystem for SSE lost
Next.js + React	plain React SPA; Vue/Svelte	team familiarity (concept notes); SSR for report pages; ecosystem	heavier than vanilla SPA
MapLibre (2D-first)	Cesium-first; Leaflet	georeferenced raster overlays, performant, open; 3D deferred	3D wow-factor deferred to ADV
PostGIS	vector DB; plain files	spatial queries (intersection, area) are core; evidence needs relational integrity	operational overhead vs. SQLite
Rasterio first, GDAL beneath	GDAL CLI-first; xarray-first	Pythonic windowed I/O; GDAL still available for conversions	some ops need GDAL CLI anyway
REST + WebSocket	REST-only; gRPC	REST for CRUD/upload; WS for live trace streaming (demo-critical)	two channels to maintain
Modular monolith first	microservices from day one	SIH timeline; single deploy; split only at GPU contention	scaling requires later refactor
Local/open-weight VLM	API VLM (GPT-4V-class)	offline judging, no per-call cost, fine-tunable (LoRA), data stays local	GPU requirement; quality below frontier APIs
LoRA/QLoRA adaptation	full fine-tuning; no adaptation	fits single GPU; proven for RS-VLMs (RS-LLaVA)	adapter quality ceiling
Synchronous core + async heavy jobs	fully async everything	simple mental model; CD/segmentation run as jobs with WS progress	API design slightly hybrid
Late fusion for optical-SAR	early fusion; learned cross-modal net	auditable per-sensor evidence; robust to mis-registration; no paired-training burden	may underuse cross-modal signal
Index engine as deterministic tool	let VLM estimate indices	closed-form, exact, auditable, free	none meaningful

Decision	Alternatives	Reason for selection	Trade-off
Evidence store (PostGIS)	trace-only logging	evidence recall queries (“show the region”) need queryable geometry	schema design effort
CPU fallback via ONNX	GPU-only	demo resilience on judges’ hardware	reduced throughput
Docker Compose deployment	Kubernetes; bare metal	reproducible machine demo	single- no horizontal scaling
English-only UI at SIH	multilingual	scope control	limits field usability [Future]

24 Risk Register

Severity = Probability $\times$ Impact (H/M/L). Every risk has a detection signal, a mitigation, and a demo-safe fallback.

Table 13: Risk register.

Risk	Prob.	Imp.	Sev.	Detection	Mitigation	Fallback
VLM hallucination	H	H	H	answers without evidence links	evidence-first rendering; constrained prompts; validation checks	show structured results without prose
Poor spatial reasoning	H	M	H	wrong counts/locations in tests	route counting/localisation to tools, not VLM	tool-computed answers only
Incorrect grounding	M	H	H	low IoU on grounding eval	RS-specific grounding model; threshold + “no confident region” state	box from specialist detector instead
Cloud cover	H	M	H	SCL/cloud-mask coverage stats	mask + warn; suggest alternate date/SAR	report “insufficient clear pixels”
SAR complexity	M	M	M	SAR misinterpretation in review	modality-aware explanations; calibrated preprocessing	restrict SAR to evidence display
Optical/SAR misalignment	M	H	H	registration residual above threshold	co-registration stage with abort threshold	per-sensor answers, no fusion
Temporal misregistration	M	H	H	edge-aligned false change in masks	sub-pixel registration check; residual map in trace	require better pair; warn user
Spatial resolution mismatch	M	M	M	grid alignment warnings	harmonise to analysis grid; document resampling	analyse at coarsest common grid

Risk		Prob.		Imp.	Sev.	Detection	Mitigation		Fallback
Domain shift (models vs. AOI)		H	M		H	confidence drop on new regions	confidence flagging; LoRA adaptation on target-like data		label outputs “unverified region”
Dataset limitations		M	M		M	benchmark-vs-field gap	multi-dataset eval; field-scene test set		demo on validated AOIs
Insufficient training data		M	M		M	poor fine-tune metrics	use pretrained specialists; small LoRA sets		zero-shot/pretrained only
GPU limitations		M	H		H	OOM errors; latency spikes	4/8-bit quantisation; tiling; small specialists		CPU fallback (slow mode)
Inference latency		M	M		M	per-stage latency in traces	async jobs + progress streaming; caching		pre-computed demo scenes
API costs (if API VLM used)		L	M		L	billing dashboard	open-weight default		n/a (local default)
Large raster memory		H	M		H	memory profiling	windowed reads; tiling; dtype discipline		downsampled analysis copy
Unsupported file formats		M	L		L	ingestion validation errors	format sniffing + clear errors; GDAL conversion		ask user for GeoTIFF/PNG
CRS errors		M	H		H	bounds/area sanity checks	EPSG validation; reprojection service		reject with explanation
False change detection		H	M		H	seasonal/edge artefacts in tests	cloud mask; seasonal-pair warning; min-area filter		present change as “candidate”
Confidence mis-calibration		M	M		M	reliability plots on eval set	temperature scaling; conservative aggregation		qualitative badges (low/med/high)
Explainability gaps		L	M		L	user testing feedback	evidence explorer; trace UI		manual walk-through in demo
Deployment complexity		M	M		M	setup time on fresh machine	Docker Compose; scripted setup		pre-built demo bundle
Integration complexity		H	M		H	end-to-end test failures	typed module interfaces; contract tests		staged demo (per-module)
Scope creep		H	H		H	feature growth in standups	MVP/SIH/ADV/FUT tiers (this document); freeze after Phase 5		Tut to MVP script

Risk	Prob. Imp. Sev.			Detection	Mitigation	Fallback
Team skill gaps	M	H	H	learning-roadmap checkpoints	role-based learning matrix; pair programming	reassign scope to strengths
Demo-day environment failure	L	H	M	rehearsal on target hardware	offline recorded video; bundle backup	video + pre-computed scenes

Part V

Execution Plan

25 Feasibility Analysis

25.1 Can all 40 features be built?

No — not within an SIH cycle with a student team and one GPU, and attempting it would produce forty shallow features instead of a credible system. The tiering below optimises for **technical credibility + working integration + strong demonstration**, not feature count.

25.2 MVP — the smallest system that clearly solves the problem

GeoTIFF ingestion + metadata (#15,16), tiling (#17), geospatial engine (#18), index engine (added), 2D map + exports (#19,39), RS-VLM VQA/captioning (#3–5), query router + registry + traces (#1,25,26,29), land-cover from segmentation (#9 via pretrained model). **Proof:** upload an optical scene, ask “what is here / where is the water / which regions have high NDVI”, get evidence-linked answers with a visible trace.

25.3 SIH demo — the strongest realistic set

MVP + grounding (#6), detection and segmentation overlays (#7,8), bi-temporal change detection with quantified, localised answers (#10), change description (#11), SAR ingestion + per-sensor analysis (#13), evidence graph + confidence (#27,28), ask-this-region (#22,23), report generation + persistence (#38,40). **Demo script:** the concept notes’ built-up-change dialogue, live, ending in a PDF report with embedded trace ID.

25.4 Advanced version — if time/resources permit

Change VQA model (#12), optical-SAR fusion view (#14), timeline scrubber (#21), autonomous investigation macro (#24), model observability page (#30), missions and histories (#33,36,37), 3D Earth (#20).

25.5 Research / future — do not attempt during the hackathon

Continuous monitoring + alerts (#34,35) — needs automated data acquisition, scheduling, and storage budgets; voice (#2,32) — ASR adds a failure layer with no analytical value; benchmark/research mode (#31) — valuable only after the system exists; end-to-end text-to-mask RES and learned cross-modal change networks — open research problems; multi-temporal (>2) reasoning and hyperspectral support.

Reasoning summary

The cut lines follow three rules: (i) *no feature without its dependency chain* — monitoring needs missions, storage, scheduling, and acquisition automation that SIH cannot supply; (ii) *no unverifiable claims* — features whose output cannot be checked live (free-form change VQA without grounding) are subordinated to checkable ones; (iii) *demo value per engineering day* — 3D Earth and voice are high-cost, low-substance for judges evaluating an ISRO problem statement.

26 Build Order

Dependency-aware sequence (Figure 1 gives the feature-level view).

Phase 0 — Fundamentals. Remote sensing + Python + Rasterio/GDAL basics; team reads Section 3; hello-world: load S-2 GeoTIFF, print metadata, render RGB.

Phase 1 — Raster engine. Ingestion, metadata, CRS, tiling, index engine, basic viewer overlays. *Gate:* NDVI map of a Sentinel-2 tile rendered in the UI.

Phase 2 — Specialist models. Segmentation (pretrained → fine-tuned), detection, change detection (LEVIR-CD-trained), registration stage. *Gate:* change mask + area stats computed for a test pair.

Phase 3 — VLM. RS-VLM serving (quantised), VQA + captioning over tiles. *Gate:* single-image Q&A in chat.

Phase 4 — Query router. Intent classifier + deterministic routing over the six core intents. *Gate:* routing accuracy $\geq 95\%$ on labelled query set.

Phase 5 — Tool calling. Router dispatches specialist tools; structured results flow back. *Gate:* “show unhealthy vegetation” runs index engine end-to-end.

Phase 6 — Evidence layer. Grounding links, evidence store, confidence badges, execution-trace UI. *Gate:* every claim in a session is clickable to pixels.

Phase 7 — Full UI. Map + chat + evidence explorer + investigation workspace.

Phase 8 — Integration. Bi-temporal and cross-modal query paths end-to-end; report generation. *Gate:* full demo script passes three consecutive runs.

Phase 9 — Evaluation. Benchmark runs (VRSBench subset, LEVIR-CD, CDVQA subset, routing set); latency/memory profiling; fix worst offenders.

Phase 10 — Demo hardening. Offline bundle, pre-computed fallback scenes, backup video, rehearsed script, freeze.

27 Learning Roadmap

Ordered by dependency, not popularity. **Must learn** vs. **nice to know** is marked per topic; each topic lists what to learn, what to skip, a practical exercise, and the SatQuery AI feature it unlocks.

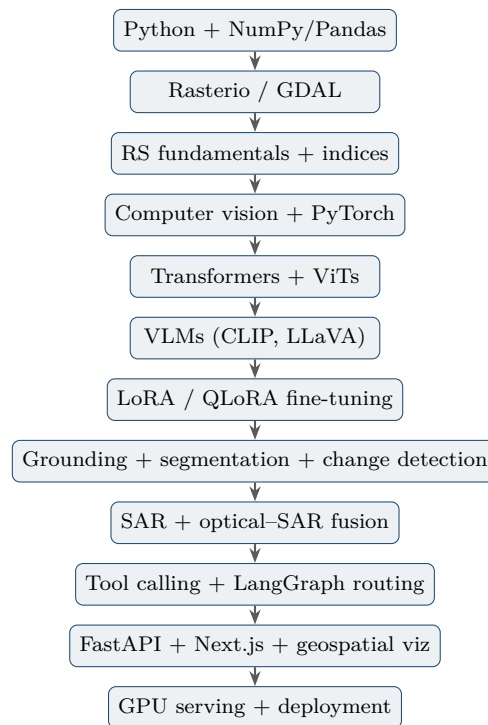


Table 14: Learning roadmap detail (M = must learn, N = nice to know).

Topic	Learn / skip	Practical exercise	Unlocks	Priority
Python + NumPy/Pandas	learn: typing, async basics, array ops; skip: metaprogramming	raster statistics script	everything	M
Math for ML	learn: linear algebra, probability, gradients; skip: proofs-heavy calculus	implement cosine similarity, IoU	model literacy	M
ML/DL fundamentals	learn: train/val/test, overfitting, transfer learning; skip: exotic optimisers	fine-tune a classifier on EuroSAT	all model work	M
RS fundamentals	learn: spectrum, 4 resolutions, sensors, SAR basics; skip: orbital mechanics	band-math notebook (NDVI/NDWI/NDBI)	index engine, SAR features	M
Geospatial computing	learn: CRS, geotransforms, reprojection, tiling; skip: datum theory deep-dive	GeoTIFF→tiles→reproject engine	reproject engine	M

Topic	Learn / skip	Practical exercise	Unlocks	Priority
Computer vision	learn: classification/detection/segmentation distinctions; skip: GANs	train U-Net on satism/grounding dataset	seg + detection	M
Transformers + ViT	learn: attention, patches, positional encoding; skip: kernel derivations	visualise attention maps	VLM work	M
VLMs	learn: CLIP alignment, LLaVA architecture, instruction tuning; skip: pre-training at scale	run GeoChat/RS-LLaVA inference	VQA/captioning	M
LoRA/QLoRA	learn: PEFT workflow, quantisation; skip: full fine-tuning	LoRA-tune on VRSBench subset	domain adaptation	M
Grounding	learn: REC/RES, boxes vs. masks; skip: 3D grounding	Grounding-DINO + SAM demo on RS tile	grounding feature	M
Segmentation	learn: U-Net/SegFormer, IoU/Dice; skip: panoptic everything	fine-tune SegFormer on LoveDA	land-cover analysis	M
Change detection	learn: Siamese nets, BIT/ChangeFormer, registration; skip: video CD	train + eval on LEVIR-CD	change features	M
SAR	learn: backscatter, VV/VH, speckle, distortions; skip: InSAR processing	visualise S-1 vs S-2 same AOI	SAR + fusion	M
Optical-SAR fusion	learn: early/late/feature fusion trade-offs; skip: building a novel fusion net	late-fusion water/built-up experiment	cross-modal lab	N (ADV)
Agentic AI / LangGraph	learn: state machines, tool calling, structured outputs, traces; skip: open-ended multi-agent	deterministic router over 6 intents	routing + traces	M

Topic	Learn / skip	Practical exercise	Unlocks	Priority
Evaluation	learn: IoU/mAP/F1, CIDEr/BLEU limits, calibration; skip: exotic NLG metrics	build per-layer eval harness	evaluation framework	M
FastAPI + back-end	learn: async, WebSockets, background jobs; skip: GraphQL	upload→process→stack API	stack	M
Next.js + maps	learn: React, MapLibre overlays, streaming UI; skip: WebGL shader programming	map with mask/change overlays + chat	frontend	M
CesiumJS 3D	learn: only if ADV tier reached	3D AOI viewer	3D Earth	N (ADV)
GPU serving + MLOps	learn: quantisation, ONNX, Docker, monitoring basics; skip: Kubernetes	one-command GPU inference service	deployment	M
ASR / voice	skip for SIH	—	voice	N (FUT)
Geospatial foundation models	learn: capabilities/limitations; skip: pretraining	compare Re-moteCLIP vs CLIP zero-shot	embedding sup-port	N

28 Research Gaps and SatQuery AI’s Potential Novelty

28.1 What existing systems do not combine effectively

The survey shows clear islands of capability [Fact]: RS-VLMs converse but do not quantify; change detectors quantify but do not converse; grounding models localise but not over validated specialist outputs; fusion models combine sensors but not under natural-language task routing; monitoring platforms exist but without conversational, evidence-linked interaction. No surveyed system combines: **natural-language interaction + multimodal RS + specialist analysis tools + task routing + grounding + temporal reasoning + optical-SAR reasoning + evidence-backed explanations** in one auditable workflow.

28.2 Novelty classification

Table 15: Honest novelty assessment.

Category	SatQuery AI elements
Established research	RS captioning/VQA/grounding models; CD architectures; fusion methods; indices
Existing capabilities (integrated, not invented)	GeoChat-class dialogue; LEVIR-CD-class change models; MapLibre/Cesium visualisation
Engineering integration (our main contribution)	deterministic router + specialist registry + evidence graph + provenance over RS tools
Potentially novel combination	evidence-first change explanation: change masks/statistics verbalised with per-pixel traceability, plus dual optical–SAR evidence in one answer
Genuinely research-worthy (post-SIH)	calibrated confidence for routed RS pipelines; grounded change QA with quantified answers (cf. CDQAG); cross-modal evidence fusion with auditability

We do not claim algorithmic novelty; we claim a defensible, verifiable *systems* contribution [Proposal].

29 Competitive / Research Comparison

Table 16: Capability comparison (● = strong/native, ○ = partial/limited, – = absent).

Capability	Traditional GIS	RS-ML pipelines	Generic VLM	RS-VLM	SatQuery AI
Natural-language interaction	–	–	●	●	●
Optical imagery analysis	●	●	○	●	●
SAR analysis	●	○	–	○	○
Optical–SAR fusion	○	○	–	○	○
Temporal / change detection	○	●	–	○	●
VQA	–	–	●	●	●
Grounding / localisation	○	○	○	●	●
Segmentation	○	●	–	○	●
Spectral indices	●	●	–	–	●
Tool calling / routing	–	–	○	–	●
Evidence localisation	○	○	–	○	●
Provenance / audit trail	○	○	–	–	●
Explainability	○	○	○	○	●
Multi-step analysis	● (manual)	○	○	○	●
Quantified answers (area, %)	●	●	–	–	●

This matrix is deliberately conservative: SatQuery AI scores ● only where the design binds the capability to verifiable tooling, and ○ for SAR/fusion where the SIH tier provides per-sensor evidence rather than learned deep fusion. The defensible claim is not “better than RS-VLMs at VQA” but “the only column where routing, evidence, provenance, and quantification are all native” [Proposal].

30 Final End-to-End System Blueprint

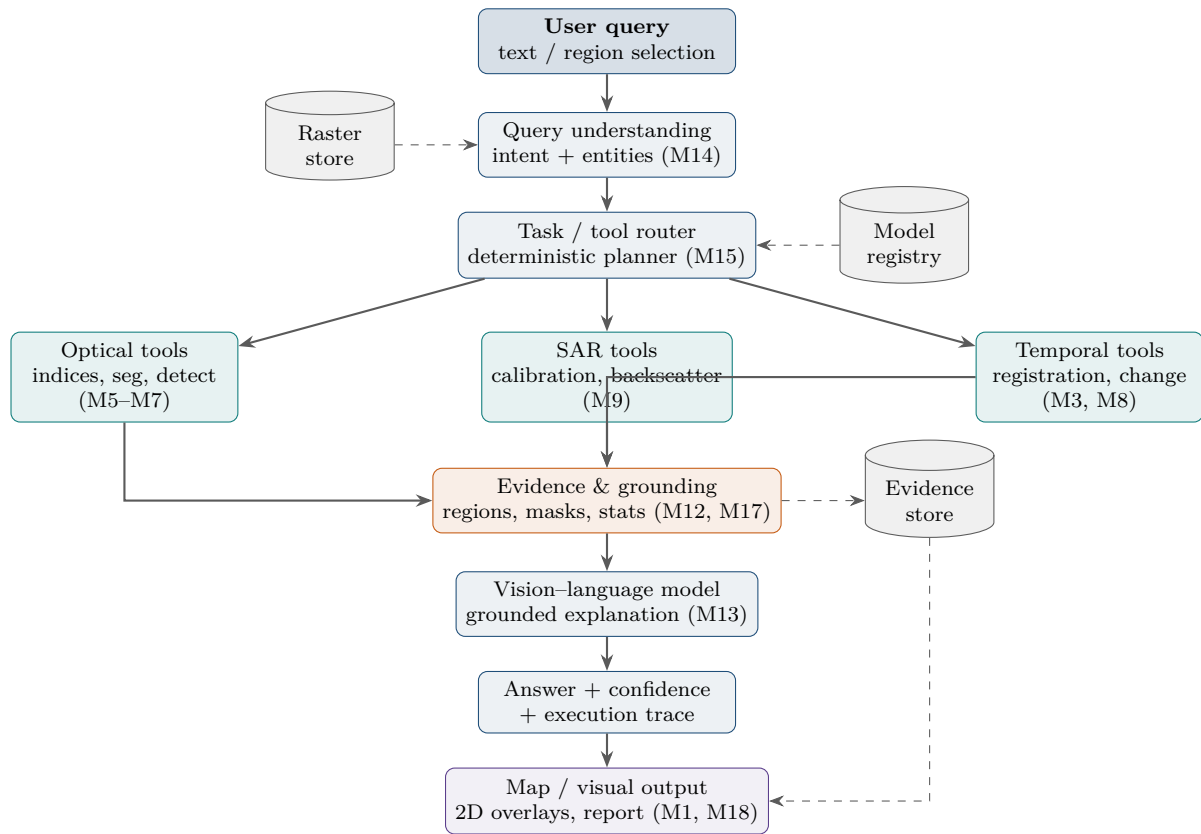


Figure 2: End-to-end architecture. Solid arrows: execution flow; dashed: data/registry dependencies. Every arrow writes to the provenance log.

The blueprint instantiates the document’s thesis: language enters at the top, specialist computation happens in the middle, evidence binds the two, and everything is logged. A judge walking the demo sees, in order: query → stated plan → live tool execution → localised evidence → quantified answer → trace → exportable report — the complete argument for SatQuery AI in one interaction.

Part VI

References

References

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